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Research Report

# Treatment-induced neural reorganization in aphasia is language-domain specific: Evidence from a large-scale fMRI study

*Elena Barbieri* [a,b,*,1], *Cynthia K. Thompson* [a,b,c,1], *James Higgins* [a,d], *David Caplan* [a,e], *Swathi Kiran* [a,f], *Brenda Rapp* [a,g] and *Todd Parrish* [a,d]

[a] *Center for the Neurobiology of Language Recovery, Northwestern University, 2240 Campus Drive, Evanston, IL 60208, USA*
[b] *Department of Communication Sciences and Disorders, School of Communication, Northwestern University, 70 Arts Circle Drive, Evanston, IL 60208, USA*
[c] *Department of Neurology, Feinberg School of Medicine, Northwestern University, 303 East Chicago Avenue, Chicago, IL 60611, USA*
[d] *Department of Radiology, Feinberg School of Medicine, Northwestern University, 737 N Michigan Avenue, Chicago, IL 60611, USA*
[e] *Department of Neurology, Massachusetts General Hospital, Harvard Medical School, 15 Parkman Street, Boston, MA 02114, USA*
[f] *Department of Speech, Language, And Hearing, College of Health & Rehabilitation, Boston University, 635 Commonwealth Avenue, Boston, MA 02215, USA*
[g] *Department of Cognitive Science, Krieger School of Arts & Sciences, Johns Hopkins Univeristy, 3400 N Charles Street, Baltimore, MD 21218, USA*

ARTICLE INFO

*Article history:*
Received 1 December 2021
Reviewed 2 May 2022
Revised 14 August 2022
Accepted 29 November 2022
Action editor Daniel Mirman
Published online 17 December 2022

*Keywords:*
Language treatment
Neuroplasticity
Sentence production
Sentence comprehension
Anomia

ABSTRACT

Studies investigating the effects of language intervention on the re-organization of language networks in chronic aphasia have resulted in mixed findings, likely related to—among other factors—the language function targeted during treatment. The present study investigated the effects of the type of treatment provided on neural reorganization. Seventy individuals with chronic stroke-induced aphasia, recruited from three research laboratories and meeting criteria for agrammatism, anomia or dysgraphia were assigned to either treatment ($N = 51$) or control ($N = 19$) groups. Participants in the treatment group received 12-weeks of language intervention targeting sentence comprehension/production, naming, or spelling. At baseline and post-testing, all participants performed an fMRI story comprehension task, with blocks of auditorily-presented stories alternated with blocks of reversed speech. Participants in the treatment, but not control, group significantly improved in the treated language domain. FMRI region-of-interest (ROI) analyses, conducted within regions that were either active (or homologous to active) regions in a group of 22 healthy participants on the story comprehension task, revealed a significant increase

* *Corresponding author*. Aphasia and Neurolinguistics Research Laboratory Department of Communication Sciences and Disorders Northwestern University 2240 Campus Drive Evanston, IL 60208 USA
E-mail address: elena.barbieri@northwestern.edu (E. Barbieri).
[1] Joint first authors.
https://doi.org/10.1016/j.cortex.2022.11.008

Spelling
Right hemisphere

in activation from pre-to post-treatment in right-hemisphere homologues of these regions for participants in the sentence and spelling, but not naming, treatment groups, not predicted by left-hemisphere lesion size. For the sentence (but not the spelling) treatment group, activation changes within right-hemisphere homologues of language regions were positively associated with changes in measures of verb and sentence comprehension. These findings support previous research pointing to recruitment of right hemisphere tissue as a viable route for language recovery and suggest that sentence-level treatment may promote greater neuroplasticity on naturalistic, language comprehension tasks, compared to word-level treatment.



## 1. Introduction

Recovery of language functions following stroke is a non-linear process, in which greater changes in language performance are observed in the early stages (days or weeks after the stroke), and less improvement occurs as recovery enters the chronic stage (from months to years after stroke, see Kiran & Thompson, 2019; Hillis, 2007, for reviews). Studies of language recovery in chronic aphasia have demonstrated that language intervention results in improved language functions and changes in neural activation and connectivity patterns (as measured by functional magnetic resonance imaging (fMRI)) even years after stroke (Allen, Mehta, Andrew McClure, & Teasell, 2012; Crosson et al., 2019; also see Hartwigsen & Saur, 2019; Kiran & Thompson, 2019 and Thompson, 2019, for reviews).

Multiple factors play a role in recovery of language (see reviews in Fridriksson & Hillis, 2021; Kiran & Thompson, 2019; Watila & Balarabe, 2015): intrinsic factors are related to individuals' demographics (e.g., age), language profile (e.g., type and severity of aphasia) and stroke characteristics (e.g., size and location of the lesion, integrity of white matter tracts); whereas, extrinsic factors refer to, for example, the type of the language intervention received. All these factors have been shown to influence the extent to which language abilities improve over time (e.g., Hope, Seghier, Leff, & Price, 2013; Lazar et al., 2010; Maas et al., 2012; Pedersen, Vinter, & Olsen, 2004; Plowman, Hentz, & Ellis, 2012; Spell et al., 2020). With regard to the effects of treatment on the reorganization of language networks, the literature is mixed as to whether tissue in the ipsilesional (left) hemisphere and/or contralesional (right hemisphere) is best suited to support recovery (see Kiran & Thompson, 2019, for review). Clearly, intrinsic factors, including lesion characteristics caused by the stroke, affect this. Large left-hemisphere lesions, for example, may preclude left-hemisphere recruitment and, hence, promote right-hemisphere recruitment (Heiss & Thiel, 2006; Heiss, Thiel, Kessler, & Herholz, 2003; Skipper-Kallal et al., 2017; Sebastian & Kiran, 2011), which some argue is associated with negative outcomes (Fernandez et al., 2004; Marcotte et al., 2012; Postman-Caucheteux et al., 2010). Some studies also suggest that damage to specific regions within the left hemisphere fosters right hemisphere recruitment, such as the left inferior frontal gyrus (IFG, Turkeltaub, Messing, Norise, & Hamilton, 2011; Sims et al., 2016), and that damage to left temporal regions may result in shifts in activation to right temporal regions (Sims et al., 2016).

The extant literature also suggests that the type of the language intervention shapes neural recovery patterns in chronic aphasia (see Gainotti, 2015; Kiran & Thompson, 2019, for reviews) and that regions recruited include those within domain-specific language networks (e.g., Abel, Weiller, Huber, Willmes, & Specht, 2014; Bonilha, Gleichgerrcht, Nesland, Rorden, & Fridriksson, 2016; Fridriksson, Richardson, Fillmore, & Cai, 2012; Johnson, Meier, Pan, & Kiran, 2019; Kiran, Meier, Kapse, & Glynn, 2015). Semantically-based treatment approaches (e.g., typicality-based semantic feature analysis, see Johnson et al., 2019; Kiran et al., 2015) result in significant post-treatment recruitment of right (mostly frontal) as well as left hemisphere regions; whereas phonologically-based treatment is more likely to result in changes within the left hemisphere (Rochon et al., 2010; van Hees, McMahon, Angwin, de Zubicaray, & Copland, 2014). Notably, these findings are in line with studies of healthy participants who show bilateral temporal activation associated with semantic (Binder, Westbury, McKiernan, Possing, & Medler, 2005; Kielar, Deschamps, Jokel, & Meltzer, 2016; Vigneau et al., 2011; Wright, Stamatakis, & Tyler, 2012; see Gainotti, 2013, for a review) and left-lateralized activation for phonological (Paulesu et al., 2000; Vigneau et al., 2011; Zatorre, Meyer, Gjedde, & Evans, 1996) processing. Similarly, recovery of spelling has been linked to left-hemisphere changes in functional activation (De Marco, Wilson, Rising, Rapscak, & Beeson, 2018; Purcell et al., 2017), functional connectivity (Tao & Rapp, 2020), or neural differentiation (as measured by Local H-Reg, Purcell, Wiley, & Rapp, 2019) in the left fusiform gyrus and the left IFG, regions recruited for normal spelling (Purcell, Turkeltaub, Eden, & Rapp, 2011). In contrast, research has suggested that right-hemisphere resources may be recruited more often for comprehension than for production (Crinion and Price, 2005; Gajardo-Vidal et al., 2018; Zaidel, 1976). In keeping with this, right-hemisphere recruitment has been frequently reported in studies of recovery from sentence comprehension deficits within regions homologous to or overlapping with those engaged by healthy listeners (Barbieri, Mack, Chiappetta, Europa, & Thompson, 2019; Thompson, den Ouden, Bonakdarpour, Garibaldi, & Parrish, 2010; Wierenga et al., 2006; see Walenski, Europa, Caplan, & Thompson, 2019 for a meta-analysis showing that sentence comprehension engages a bilateral network). Among these studies, two (Barbieri et al., 2019; Thompson et al., 2010) investigated the effects of Treatment of Underlying Forms (TUF, Thompson & Shapiro, 2005), a linguistically-based language intervention focused on restoring the processes of thematic role assignment and thematic-syntactic mapping that underlie complex sentence processing, and found treatment-induced upregulation of activation (during fMRI comprehension tasks) in several right hemisphere regions that was positively correlated with treatment outcomes and changes in on-line sentence processing.

Taken together, these findings support one principle for promoting neuroplasticity of language networks in aphasia: *specificity rebuilds language networks* (Kiran & Thompson, 2019; Kleim & Jones, 2008), suggesting that the neural pathways underlying recovery of language are a function of the treatment provided. However, only a few studies have addressed this question, for example comparing different approaches to treatment of naming deficits (e.g., Abel, Weiller, Huber, & Willmes, 2014; Kurland, Pulvermüller, Silva, Burke, & Andrianopoulos, 2012; van Hees et al., 2014), but have used experimental designs that did not allow the effects of each treatment on neural activation to be evaluated (see van Hees et al., 2014).

Furthermore, the task employed to evaluate treatment-induced changes in activation is also likely to play a role in the patterns of neural recovery observed on fMRI. As most of the literature on aphasia recovery has focused on the effects of treatments for anomia, studies have mostly employed word-level fMRI tasks, such as overt naming (Abel et al., 2014; Fridriksson et al., 2012; Kiran et al., 2015; Meinzer et al., 2008; Menke et al., 2009; Raboyeau et al., 2008; van Hees et al., 2014), category-based word generation (Crosson et al., 2005), semantic and/or rhyme judgments (Rochon et al., 2010). Only a handful have employed tasks targeting language processes that extend beyond the level of single words (e.g., Barbieri et al., 2019; Cherney & Small, 2006; Thompson et al., 2010; Wierenga et al., 2006). Among these, only one (Cherney & Small, 2006) used a naturalistic task (passive comprehension of stories) to evaluate treatment-induced changes in activation. Because fMRI paradigms are typically designed to target the same language functions targeted during treatment, the effects of the treatment type and fMRI task paradigm are often impossible to disentangle. In the present study, we overcome this limitation by evaluating the effects of three types of treatment (provided for different language functions) using a common, naturalistic, story comprehension task.

Another topic that has received increased attention in the literature on aphasia recovery is the engagement of domain-general functions and associated neural mechanisms to support language recovery. Studies have shown that pre-treatment accuracy on tests of domain-general functions, including attention, predict treatment outcome in participants with aphasia (Lambon Ralph, Snell, Fillingham, Conroy, & Sage, 2010; Harnish & Lundine, 2015; see Villard & Kiran, 2017, for a review). In addition, neuroimaging research suggests that regions supporting domain-general functions such as attention and executive control are engaged during treatment (Brownsett et al., 2014; Geranmayeh, Brownsett, & Wise, 2014; Geranmayeh, Chau, Wise, Leech, & Hampshire, 2017), and are associated with improved sentence comprehension and improved on-line integration of semantic and syntactic information during processing of complex sentences (Barbieri et al., 2019). In this respect, domain-general regions have been proposed to exert an overarching effect on language networks, thereby supporting recovery of language functions (Geranmayeh, Brownsett, & Wise, 2014). In keeping with this hypothesis, inhibition of domain-general regions using low-frequency transcranial magnetic stimulation (TMS) impairs language learning in healthy individuals (Sliwinska et al., 2017).

### 1.1. *The current study*

In the present study, we compared the effects of treatment targeting different language domains (sentence processing, naming, spelling) on neural plasticity in a large group of individuals with chronic stroke-induced aphasia, using an fMRI auditory story comprehension task, a common task that was administered to all study participants prior to and following treatment. In healthy individuals, listening to short stories activates a large network including the superior temporal gyri (STG), the left fusiform gyrus, the left middle temporal gyrus (MTG) and the left frontal cortex encompassing the IFG, the superior frontal gyrus (SFG), and premotor and motor areas (Crinion & Price, 2005; Skipper, Nusbaum, & Small, 2005; Wilson, Molnar-Szakacs, & Iacoboni, 2008; Yarkoni, Speer, & Zacks, 2008). Notably, this network largely overlaps with regions active during sentence comprehension tasks, as demonstrated by Yarkoni et al. (2008), who found that reading narratives and single sentences yielded activation in the same areas, with the exception of the bilateral dorsomedial prefrontal cortex activation, which was exclusively associated with narrative-level comprehension. Aside from reliably activating a large left fronto-temporal network in healthy speakers, the use of an auditory story comprehension task allows for investigation of language processing in a "naturalistic" condition and is suitable for individuals with any level of aphasia severity.

The study aimed to 1) evaluate behavioral and activation changes (from pre-to post-treatment) in three groups of participants with aphasia who underwent treatment for different language impairments (sentence processing, naming, spelling) and in a group who did not receive treatment (i.e., natural history group); 2) investigate the contribution of domain-general regions to language recovery; and 3) examine the relation between changes in activation and changes in language performance.

For the first aim, based on previous research (Barbieri et al., 2019; Gilmore, Meier, Johnson, & Kiran, 2019; Johnson et al., 2019; Kiran et al., 2015; Purcell et al., 2019; Thompson et al., 2010; Wiley & Rapp, 2019), we expected all types of treatment to result in increased accuracy on domain-specific language measures from pre-to post-testing. For the common story comprehension fMRI task, we anticipated that, due to overlaps in neural activation associated with sentence processing and narrative comprehension (Yarkoni et al., 2008), individuals who received treatment for sentence processing would show significant upregulation (i.e., increased activation from pre-to post-treatment) within language regions that were active in healthy participants on the same story comprehension task, or in right hemisphere homologues of these regions. Additionally, because the naming treatment exploited word-level semantic knowledge (see Gilmore et al., 2019; Kiran et al., 2015), a process engaged for understanding short stories, we hypothesized that significant activation increases (i.e., greater activation post-vs. pre-treatment) would also be forthcoming for individuals who received treatment for naming. However, we predicted no activation changes for participants who underwent treatment for spelling deficits or those who received no treatment (natural history group). As for the second aim of the study, we predicted that, if treatment promotes recruitment of domain-general regions to support language processes (in conjunction with language-specific regions), groups who show significant treatment-induced activation changes in language regions would also exhibit increased activation within domain-general regions that were active in healthy participants during the story comprehension task, or in homologues of such regions. Upregulation of domain-general regions was expected to be greater for participants in the sentence production/comprehension treatment group than for those in the naming and the spelling treatment groups, based on the evidence that allocation of attentional resources during language tasks may be a function of task demands (Jongman, Roelofs, & Meyer, 2015; Nozari, Trueswell, & Thompson-Schill, 2016; Whitney, Kirk, O'Sullivan, Lambon Ralph, & Jefferies, 2012), thereby implying that attentional resources are more likely recruited by sentence-level (vs. word-level) treatment. Finally, with respect to the third aim of the study, we expected that changes in activation within both language and domain-general regions would be positively associated with changes in language performance, namely, with increased performance on tests assessing word and sentence comprehension.

## 2. Materials and methods

We report how we determined our sample size, all data exclusions, all inclusion/exclusion criteria, whether inclusion/exclusion criteria were established prior to data analysis, all manipulations, and all measures in the study.

### 2.1. *Participants*

Individuals with aphasia were recruited from three research centers: the Aphasia and Neurolinguistics Research Lab at Northwestern University (Evanston, IL), the Aphasia Research Lab at Boston University (Boston, MA), and the Cognitive and Brain Sciences Lab at Johns Hopkins University (Baltimore, MD). Across research centers, a total of 100 individuals with aphasia were evaluated for participation in the study. Of these, 70 met the inclusion criteria for the study: (1) presence of a single left-hemisphere stroke at least 12 months prior entering the study and affecting at least .1% of the cortical surface in the left hemisphere (i.e., individuals with exclusively subcortical lesions were not included), (2) monolingual native English speakers, (3) between 35 and 80 years old, (4) high-school educated or higher, (5) aphasia severity (as measured by the Western Aphasia Battery-Revised Aphasia Quotient—WAB-AQ; Kertesz, 2007) greater than 45 but lower than 95, (6) no history of developmental, psychiatric or neurological disorders, other than the aphasia, (7) normal or corrected-to-normal vision and hearing, and (8) no contraindications to undergoing MRI.

All participants with aphasia underwent a language evaluation using standardized tests that were identical for all participants; in addition, a spontaneous speech sample was collected through the re-telling of Cinderella's story and analyzed using the Northwestern Narrative Language Analysis (Fromm, MacWhinney, & Thompson, 2020; Hsu & Thompson, 2018). Participants were screened at each site by a trained staff member, and their eligibility for the treatment provided at each site was determined based on their language profile. Although all participants presented with aphasic impairments that affected multiple language domains, participants recruited at Northwestern University were selected for agrammatism, which was defined by: a) evidence of impaired production and comprehension of sentences (for syntactically complex (non-canonical) more than for simple (canonical) sentences) on the *Northwestern Assessment of Verbs and Sentences* (NAVS, Thompson, 2011); b) production of less than 50% grammatical sentences on the re-telling of Cinderella's story; and c) relatively spared single-word comprehension on the *Northwestern Naming Battery* (NNB, Thompson & Weintraub, 2014). At Boston University, participants were selected for anomia based on a study-specific naming battery including natural and artificial objects that belonged to four different semantic categories (see Johnson et al., 2019, for details). Participants at Johns Hopkins University were recruited if they showed evidence of dysgraphia (defined as <80% letter accuracy) on the *Johns Hopkins University Dysgraphia Battery* (Goodman & Caramazza, 1985), as well as on selected subtests (#40 and #54) of the *Psycholinguistic Assessment of Language Processing in Aphasia* (PALPA; Kay, Lesser, & Coltheart, 1992).

A group of healthy, non-brain damaged individuals ($N = 22$, all monolingual native English speakers, with normal or corrected-to-normal vision and hearing, without history of developmental, psychiatric or neurological disorders, and without any contraindications for undergoing MRI) were recruited from the Chicago area. All participants provided written informed consent according to Northwestern University Institutional Review Board (IRB) policies. The study was conducted in accordance with the rules established by the Declaration of Helsinki for experiments with human subjects.

Following baseline language assessment, participants were randomly assigned to the Treatment, or the Natural History (i.e., no treatment) group. Randomization of participants to the Treatment or Natural History groups occurred independently at each recruitment site and was carried out by a staff member other than the person who performed the language assessments. The Principal Investigator(s) at each recruitment site were not blinded to allocation of participants. Across recruitment sites, a total of 51 participants were randomized to receive treatment, and 19 to serve as controls (natural history group). Treatment groups received language interventions targeted to their language deficits, i.e., sentence production and comprehension (agrammatic group, $N = 15$), naming (anomic group, $N = 18$), or spelling (dysgraphic group, $N = 18$) treatment; whereas, participants in the natural history group ($N = 5$ agrammatic, $N = 5$ anomic, and $N = 9$ dysgraphic) received no treatment.

Seven participants dropped out prior to completing the study; in addition, one participant assigned to the sentence treatment group was excluded after developing a hematoma during the treatment period. Four additional participants were excluded because neuroimaging data did not meet quality standards. The total number of participants across sites included in data analyses was 58, distributed as follows: 12 participants received treatment for sentence production/comprehension deficits, 17 received naming treatment, 13 received spelling treatment and 16 received no treatment. Demographic information and pre-treatment language scores for all groups are provided in Table 1.

Between-group comparisons (carried out using simple regressions or Chi-square tests for categorical variables) revealed that individuals in the sentence production/comprehension treatment group were younger than all other groups (naming treatment: $t = -3.278$, $p = .0018$; spelling treatment: $t = -2.632$, $p = .011$; natural history: $t = -2.731$, $p = .0085$); in addition, individuals in the naming treatment group had on average a lower number of years of education than participants in the sentence production/comprehension treatment group ($t = -2.124$, $p = .0383$, Table 1). No significant differences were found between groups with respect to months post-onset ($F = .805$, $p = .4964$), gender ($\chi^2 = .559$, $p = .9058$) and handedness ($\chi^2 = 9.041$, $p = .1713$). Aphasia severity (WAB-AQ) and lesion size (i.e., the percentage of lesioned cortical gray matter in the left hemisphere) also did not differ across groups ($F = 1.098$, $p = .3579$ and $F = 1.497$, $p = .2258$, respectively). Results of language testing showed no differences between groups on any of the subtests of the Northwestern Naming Battery (NNB; Thompson & Weintraub, 2014) (noun production: $F = .742$, $p = .5317$; noun comprehension: $F = .946$, $p = .4247$; verb production: $F = 1.074$, $p = .3677$; verb comprehension: $F = 1.378$, $p = .2594$). Similarly, no between group differences were found on the Northwestern Assessment of Verbs and Sentences (NAVS (Thompson, 2011) tests for production of either canonical or non-canonical sentences. However, some differences in sentence comprehension were found: for canonical sentences, comprehension was significantly lower in the naming treatment group than in the spelling ($t = -2.269$, $p = .0273$) and the natural history groups ($t = -2.350$, $p = .0225$), but did not differ between the naming and sentence production/comprehension groups; for non-canonical sentences, comprehension was more impaired in the sentence than in the spelling treatment group ($t = -2.729$, $p = .0085$). In addition, as expected, individuals in the sentence treatment group showed an advantage for canonical over non-canonical sentences in production (canonical > non-canonical = 31.8%), which was greater than for the other treatment groups (naming treatment: 16.5%, spelling treatment: 13.8%), although only the difference between the sentence and spelling groups reached statistical significance ($t = 2.125$, $p = .0383$). A greater canonical advantage for the sentence treatment group than for all other treatment groups was also observed in comprehension (27.2%, against 7.7% for the naming treatment and 15.5% for

**Table 1 – Demographics and language scores for treated participants recruited from each research center, and for the natural history group.**

| | Sentence production/comprehension Treatment | | Naming Treatment | | Spelling Treatment | | Natural History | | All participants | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Mean | SD | Mean | SD | Mean | SD | Mean | SD | Mean | SD |
| *Demographics* | | | | | | | | | | |
| Age | 50.00 | 7.36 | 62.00 | 8.77 | 60.23 | 10.16 | 60.13 | 11.62 | 58.60 | 10.47 |
| Education | 16.50 | 2.43 | 14.88 | 1.93 | 16.23 | 1.42 | 16.03 | 2.18 | 15.84 | 2.07 |
| MPO | 42.33 | 30.01 | 60.00 | 53.09 | 67.85 | 41.07 | 67.44 | 55.44 | 60.16 | 47.16 |
| Gender | 4F, 8M | | 6F, 11M | | 3F, 10M | | 5F, 11M | | 18F, 40M | |
| Handedness | 9R, 3L | | 17R | | 10R, 2L, 1A | | 15R, 1L | | | |
| WABR-AQ | 70.93 | 12.96 | 68.92 | 14.49 | 75.34 | 15.45 | 76.92 | 13.16 | 72.98 | 14.09 |
| *NNB (% correct)* | | | | | | | | | | |
| Noun Production | 66.08 | 24.88 | 57.41 | 31.72 | 57.38 | 25.98 | 69.06 | 23.62 | 62.41 | 26.82 |
| Noun Comprehension | 99.17 | 2.89 | 93.53 | 12.22 | 95.38 | 6.60 | 95.00 | 9.66 | 95.52 | 9.02 |
| Verb Production | 73.08 | 20.73 | 63.00 | 29.27 | 63.62 | 28.23 | 76.06 | 18.72 | 68.83 | 24.87 |
| Verb Comprehension | 94.50 | 6.54 | 92.24 | 14.81 | 98.15 | 3.51 | 97.44 | 5.01 | 95.47 | 9.22 |
| *NAVS (% correct)* | | | | | | | | | | |
| SPPT C | 54.09 | 28.21 | 45.18 | 32.82 | 44.62 | 37.48 | 64.00 | 35.05 | 51.84 | 33.83 |
| SPPT NC | 21.27 | 22.53 | 28.65 | 29.76 | 30.77 | 30.31 | 34.27 | 35.34 | 29.20 | 29.81 |
| SPPT (C > NC) | 31.82 | 22.12 | 16.53 | 19.02 | 13.85 | 22.50 | 29.73 | 23.80 | 22.64 | 22.65 |
| SCT C | 78.33 | 12.12 | 72.12 | 21.90 | 86.62 | 18.32 | 86.21 | 13.96 | 80.57 | 18.02 |
| SCT NC | 51.17 | 15.42 | 64.41 | 19.82 | 71.08 | 18.97 | 63.44 | 17.72 | 62.90 | 18.96 |
| SCT (C > NC) | 27.17 | 24.87 | 7.71 | 14.13 | 15.54 | 17.44 | 22.88 | 15.60 | 17.67 | 18.99 |
| *Cinderella* | | | | | | | | | | |
| % grammatical sentences | 34.92 | 32.49 | 53.85 | 25.42 | 35.92 | 26.54 | 42.69 | 21.85 | 42.44 | 26.87 |

*Note.* R = right-handed; L = left-handed; A = ambidextrous. MPO = months-post-onset; WABR-AQ = Western Aphasia Battery-Revised, Aphasia Quotient; NNB = Northwestern Naming Battery; NAVS = Northwestern Assessment of Verbs and Sentences; SPPT = Sentence Production Priming Test; C = Canonical; NC = Non-canonical, SCT = Sentence Comprehension Test; SD = standard deviation.

the spelling treatment group), although only the difference between the sentence and the naming treatment groups reached statistical significance ($t = 2.884$, $p = .0056$). Production of grammatical sentences in spontaneous speech (i.e., retelling of Cinderella's story) was numerically lower in both the sentence and spelling groups compared to all other groups, but none of the differences was significant.

Notably, participants included in this study were part of a larger study cohort (funded by the Clinical Center Grant P50DC012283 awarded to Dr. Thompson). Data regarding behavioral and neural outcomes of treatment (sentence production/comprehension, naming, spelling) for participants largely overlapping with those included in the current study have been reported in previous studies (Barbieri et al., 2019; Gilmore et al., 2019; Johnson et al., 2019; Purcell et al., 2019). Studies reporting pre-to post-treatment changes in neural activation used domain specific tasks; none of them reported data from the fMRI task employed in this study.

### 2.2. Treatment

Descriptions of the language intervention protocols employed across research centers are provided elsewhere (Barbieri et al., 2019, for sentence production/comprehension; Gilmore et al., 2019, and Johnson et al., 2019, for naming; Purcell, et al., 2019, and Wiley & Rapp, 2019, for spelling). Briefly, at Northwestern University, participants in the treatment group underwent Treatment of Underlying Forms (TUF, Thompson & Shapiro, 2005), a treatment approach that uses a series of metalinguistic steps focused on verbs, verb argument structure, and thematic-syntactic mapping in active (e.g., *The man was shaving the boy*) and passive sentences (e.g., *The boy was shaved by the man*) to improve sentence comprehension and production. Participants who underwent TUF were trained on the same set of 10 passive sentences. Treatment steps included: identification of thematic roles (Agent, Theme) in target action pictures, guided construction of an active sentence describing the picture, clinician-demonstrated derivation of the target sentence structure (passive) from the base (active) form, and participant-initiated passive sentence construction. At Boston University, the participants treated for anomia underwent a typicality-based semantic feature analysis treatment in which typical (e.g., *pigeon*) or atypical (e.g., *penguin*) exemplars of a certain semantic category (e.g., *birds*) were trained. Each participant in the treatment group was randomly assigned to two different combinations of a semantic category (chosen from the following: *birds*, *clothing*, *furniture*, *vegetables* or *fruits*) and a typicality condition (typical or atypical: e.g., typical birds and atypical furniture). Participants were trained on half ($N = 36$) of the items included in each of the 2 trained categories; the other half was used to assess generalization. Treatment entailed several steps, including category sorting, auditory and written feature verification, and feature review. Finally, at Johns Hopkins University, training employed a spell-study-spell procedure for participants with dysgraphia. Treatment procedures required participants to listen, repeat, spell, copy and spell again each word from a pre-determined, individually tailored, list of words ($n = 40$). At all research centers, treatment was administered by a mix of trained research personnel and speech-language pathologists, and was provided with the same intensity and on the same treatment schedule (12 weeks, 2 sessions/week, 1.5 h/session).

### 2.3. Experimental procedures

#### 2.3.1. Language tasks

Primary outcome measures, specific for each language domain (sentence production/comprehension, naming, spelling), were developed at each research center and were administered to participants upon entry into the study and following a 12-week treatment (or natural history) period. Agrammatic participants who received sentence production/comprehension treatment performed two computerized tasks: a sentence production priming task, which measured changes in production of trained (passive) and other sentence structures (e.g., active sentences with unaccusative verbs, object-cleft sentences), and a picture verification task, which measured changes in comprehension of the same sentence types. In the sentence production priming task, participants were presented with two pictures, side-by-side on the computer screen, displaying semantically reversed transitive actions (e.g., *kissing*), with the participants (e.g., a man and a woman) depicted as Agent (i.e., the person doing the action) in one picture and Theme (i.e., the person undergoing the action) in the other. After hearing a sentence describing the picture displayed on the left, participants were asked to produce a similar sentence to describe the picture on the right. In the picture verification task, participants were asked to listen to a sentence and indicate if it matched a single action picture presented on the screen by pressing a button on the keyboard. Treatment efficacy was measured as the average change in production and comprehension accuracy on the 10 passive sentences that were targeted during treatment. The anomic participants who received naming treatment performed a picture naming task, in which they were shown pictures of objects belonging to different semantic categories (e.g., birds and clothing), each including typical (e.g., pigeon, for birds) and atypical (e.g., penguin) items, and were asked to name the object as fast as they could. Treatment efficacy was measured as the average change (from pre- to post-treatment) in the proportion of correctly named items in the two trained combinations of semantic category and typicality. Finally participants with dysgraphia who received spelling treatment were administered a spelling-to-dictation task; treatment efficacy was computed as the average change in the proportion of correctly spelled letters (i.e., letters that were accurately produced and appeared in the correct position within the target word), out of the total numbers of letters across all trained words. Additional details regarding these tasks are provided elsewhere (Barbieri et al., 2019; Gilmore et al., 2019; Johnson et al., 2019; Purcell et al., 2019; Wiley & Rapp, 2019).

#### 2.3.2. Neuroimaging

2.3.2.1. fMRI task. All participants (including those in the natural history group) performed a block design fMRI story comprehension task (Higgins et al., 2020; presented using E-Prime version 2.0; Psychology Software Tools, Pittsburgh, PA) at two time points, i.e., pre- and post-testing. The task was administered over two runs, for a total of four runs across time points. Each run included 8 blocks (duration = 24s/each) of short stories, 4 of which were presented auditorily and 4 visually, and 8 blocks of a control condition (duration = 18s/each) that consisted of either time-reversed speech[2] from the stories ($N$ = 4, serving as a low-level baseline for auditory stories) or degraded text ($N$ = 4, serving as a low-level baseline for visual stories).[3] Within each run, blocks were grouped by modality (auditory or visual), and the order of presentation of auditory and visual blocks was counterbalanced across runs and participants. Within each modality, experimental and control blocks were presented alternately, each preceded by a 2s fixation. A written cue (2s duration) was presented to alert the participant prior to the start of a new modality. In addition, the scan began with 8s fixation, for a total run duration of 6m22s. A 30s test run, which employed an example story not included in the task, was conducted prior to the story comprehension task, to test the participants' ability to adequately hear the auditory stimuli over the scanner noise. Although all conditions (auditory and visual) were modeled in the first-level fMRI analyses (see below), only the auditory portion of the task was analyzed further (second-level and ROI analyses). Additional details regarding the visual portion of the study can be found in Higgins et al. (2020). All auditory stories were recorded by a male native speaker of American English using Audacity, normalized in amplitude and digitally manipulated to last 24s. Stories were controlled for the number of syllables ($M = 80 \pm 2.14$), words ($M = 62.5 \pm 2.67$), nouns ($M = 19 \pm 1.51$), verbs ($M = 10.63 \pm .74$) and complex sentences ($M = 3.13 \pm .64$) and were matched across runs for the same variables (all $ps > .7$). To ensure continued attention during the task, participants were required, at the beginning of each story, to respond via button-press and indicate if they were hearing 'real' stories or not. In addition, they were told to pay attention to the stories, as they would be asked to answer yes/no questions about them at the end of the scan.

2.3.2.2. Data acquisition. Prior to data collection, imaging protocols were harmonized across sites, to ensure similar image quality and timing parameters. Harmonization was achieved by performing phantom scans at each of the three research centers, and tuning scanning parameters so that echo time and TR were consistent across sites, thereby ensuring the same sensitivity in BOLD signal detection. Structural and functional MR images were collected using 3T scanners (Siemens Trio at Northwestern University, Siemens Skyra at Boston University, and Philips Intera at Johns Hopkins University) and a 3D T1-weighted sequence for structural (TR = 2.3sec, TE = 2.91 ms, flip angle = 9°, field of view = 256 × 256, 1 mm isotropic voxel size), or a gradient-echo T2*-weighted sequence for functional (TR = 2.4 s,[4] TE = 20 ms, flip angle = 90°, 1.72 × 1.72 × 3mm voxel size, and 192 or 162 volumes) images. All images were archived on NUNDA (Northwestern University Neuroimaging Data Archive, https://nunda.northwestern.edu; Alpert, Kogan, Parrish, Marcus, & Wang, 2016) and underwent automatic pipelines for quality assurance (QA, Song, Wang, Alpert, Wang, & Parrish, 2015), following which scans that were deemed of insufficient quality were either excluded or repeated.

### 2.4. *Data analysis*

Because of the low number of participants assigned to the Natural History group at each research center, all analyses were conducted by comparing each of the three Treatment groups (sentence production/comprehension, naming, spelling) to a Natural History group that combined participants recruited at different research centers ($n$ = 16).

#### 2.4.1. *Treatment effects*

Data were analyzed using mixed-effects regressions (Baayen, Davidson, & Bates, 2008) running in R 4.1.1 (R Core Team, 2022), using the *lme4* (Bates, Mächler, Bolker, & Walker, 2014) and *lmerTest* (Kuznetsova, Brockhoff, & Christensen, 2017) packages. Since measures of treatment efficacy were unique for each treatment group, the proportion of correct responses at each time point was converted to z-scores by subtracting the pooled average (for each treatment type) from each individual's proportion correct score at each time point, and then by dividing the result by the group standard deviation (SD) at pre-testing. Mixed-effects regressions were then run using *Time* (pre-, post-testing) and *Group* (sentence processing treatment, naming treatment, spelling treatment, and natural history), as well as all their interaction, as fixed effects, and *Participant* as the only random effect. Statistical significance for fixed effects was determined by computing p-values using Satterthwaite approximation, as provided within the *lmerTest* package (Kuznetsova et al., 2017). Statistical significance of interaction terms was determined using the *anova* function, which provides a p-value reflecting the overall significance of the model term. Follow-up comparisons for fixed effects with three or more levels were carried out using the *multcomp* package (Hothorn, Bretz, & Westfall, 2008), and applying False Discovery Rate (FDR; Benjamini & Hochberg, 1995) correction for multiple comparisons.

#### 2.4.2. *Neuroimaging data*

2.4.2.1. Anatomical images. For individuals with aphasia, lesioned areas of the brain were manually drawn on the T1 image (in native space) using all the three orthogonal views simultaneously. T2/FLAIR images were used to determine the presence of additional vascular pathology. The final masks were reviewed, revised, and ultimately approved by at least two of the authors (TP, JH). Using an automated preprocessing pipeline available on NUNDA (Song et al., 2015), structural T1 images underwent enantiomorphic replacement wherein healthy tissue from the right hemisphere was mirrored into the lesioned area (Nachev, Coulthard, Jäger, Kennard, & Husain, 2008), and the resulting brains were normalized to the MNI template using the SPM DARTEL toolbox (Ashburner, 2007).

[2] Time-reversed speech preserves the phonological features of speech, without any associated meaning.

[3] Eight agrammatic participants ($N$ = 7 assigned to the treatment group, $N$ = 1 to the natural history group) underwent an earlier version of the task, which only contained blocks of auditory stories.

[4] For two agrammatic participants, functional images were collected using a 2s TR.

2.4.2.2. Functional images. Using an automated pre-processing pipeline (Song et al., 2015), blood-oxygenation-level-dependent (BOLD) data were first despiked (AFNI 3dDespike) and co-registered to the middle image (AFNI 3dvolreg). Next, motion artifact correction was performed by regressing out images that exceeded the framewise displacement (FD) threshold of 0.5 mm and a DVARS (RMS intensity differences between volumes) threshold of 50, as well as the first 3 principal components of signals extracted from white matter and cerebrospinal fluid (Behzadi, Restom, Liau, & Liu, 2007). BOLD images were then co-registered to the T1 and normalized to MNI space (using a $2 \times 2 \times 2$ mm resolution). The two transforms were concatenated and applied in one step, and a single anatomical image from each participant was used for normalization of BOLD data from all scans in order to eliminate variability caused by differing normalization parameters derived from multiple sessions. Images were then smoothed with a 6 mm FWHM kernel (AFNI 3dmerge). First-level analyses were carried out in SPM12: for healthy participants, data were modeled using a standard hemodynamic response function (HRF); however, for individuals with aphasia, time and dispersion derivatives were included in the model to account for possible effects of stroke on the HRF (Bonakdarpour, Parrish, & Thompson, 2007). Contrast images reflecting the subtraction of the auditory control condition (i.e., reversed speech) from auditory real stories (stories > control) were estimated for all participants.

2.4.2.3. Voxel-level analyses. Second-level, whole-brain group analyses, using one-sample *t*-tests, were run in SPM12 to describe activation patterns evinced by healthy participants on the fMRI story comprehension task. One-sample *t*-tests were then run on each patient group at each time point, after restricting the analysis to regions that were active in the group of healthy participants on the same task, as well as to their right-hemisphere homologues. For both analyses, T-maps were obtained by applying a $p < .001$ threshold at the voxel-level. Cluster-level correction was carried out using the family-wise error (FWE) correction provided by SPM12, at a threshold of $p < .05$ (see Eklund, Nichols, & Knutsson, 2016). Labels for active regions were derived from the Harvard–Oxford atlas (Desikan et al., 2006).

2.4.2.4. ROI analyses (aphasic participants only). Analyses were conducted by extracting the mean BOLD signal from the same regions-of-interest (ROIs) that were used to constrain the voxel-level analysis described above (i.e., from ROIs that were active in healthy participants and from their right-hemisphere homologues).

BOLD signal extracted for each individual was used as a dependent variable. Mixed-effect regression analyses were run with the aim to detect any changes in activation from pre- to post-testing within each participant group. Analyses were run on the mean BOLD signal, which was entered separately for each participant, each ROI, and each time point. *Group*, *Time*, and their interaction were included as fixed effects, and *Participant* and *ROI* as random effects. *Age, education and lesion size* (i.e., the proportion of lesioned tissue in each left-hemisphere ROI) were entered as covariates, to ensure that activation changes between time points, as well as the *Group*Time* interaction, were not driven by group differences in demographic or lesion variables. The rationale for including these covariates (and not others, such as months post-onset) was that the groups were matched for all variables except age and education; in addition, lesion size was included in view of its theoretical and historical relevance in the literature on aphasia recovery. Procedures for the computation of p-values and corrections for multiple comparisons were the same as described in 2.4.1.

In patient groups for which significant changes between time points were detected based on the ROI-based analysis and for which significant (i.e., suprathreshold) activation for the *Story* > *Control* contrast was detected on the voxel-based analysis at post-treatment, additional mixed-effect regression analyses were run to evaluate the relation between changes in activation and changes in language performance. For these analyses, the change in BOLD signal between time points (post-minus pre-treatment) was entered as the dependent variable; changes in performance on measures of word (i.e., change on the NNB noun and verb comprehension subtests) and sentence comprehension (i.e., change on the NAVS sentence comprehension subtest) were entered as fixed effects in separate analyses. *ROI* was entered as the only random effect. For these analyses, ROIs were included only if significant post-treatment activation for the *Story* > *Control* contrast was found on the voxel-based analysis. This choice was made to ensure that the relationship between upregulation of activation and changes in language scores was limited to regions for which upregulation of activation resulted in post-treatment neural response patterns to naturalistic language comprehension that were similar to those of healthy individuals (i.e., greater for real stories than for reversed speech). Procedures for the computation of p-values and corrections for multiple comparisons were the same as described in 2.4.1.

For all ROI-based analyses, the results for the full model, including all fixed and random effects detailed in this section, were reported. For post-hoc analyses, which were reported only in the event of a significant interaction, p-values were corrected for multiple comparisons using FDR correction (Benjamini & Hochberg, 1995).

## 3. Results

### 3.1. *Treatment effects*

A significant *Group*Time* interaction ($\chi^2 = 50.078$, $p < .0001$) was found, with paired comparisons showing significant differences between each of the treatment groups and the natural history group (Table 2). Post-hoc analyses revealed a significant effect of *Time*, in the direction of greater accuracy post- than pre-testing, in all the three treatment groups (sentence production/comprehension, naming, spelling, all ps < .0001, Fig. 1), whereas no change in accuracy from pre-to post-testing was found in the natural history group. No effect of *age*, *education*, or *lesion size* was found on behavioral performance.

**Table 2 – Treatment effects. Results are reported for the full model comparing pre- and post-treatment accuracy on the behavioral task measuring treatment efficacy across and between patient groups, as well as for post-hoc analyses conducted within each patient group. Variance and Standard Deviation (SD) are displayed for random effects, and beta estimates, Standard Error (SE), t-statistics and p-values are reported for all fixed effects. Reported p-values reflect FDR (Benjamini & Hochberg, 1995) correction for multiple comparisons.**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Full Model** | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | .998 | .999 | | | | | |
| Residual | .416 | .645 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | .058 | .146 | 51 | .396 | .6938 |
| Group: Sentence Prod/Comp vs. NH | | | .703 | .448 | 51 | 1.570 | .1225 |
| Group: Naming vs. NH | | | 1.146 | .392 | 51 | 2.922 | .0052 |
| Group: Spelling vs. NH | | | .782 | .416 | 51 | 1.880 | .0658 |
| Time (Post- vs. Pre-treatment) | | | 1.712 | .121 | 54 | 14.141 | <.0001 |
| Age | | | −.004 | .017 | 51 | −.243 | .8089 |
| Education | | | −.037 | .077 | 51 | −.477 | .6355 |
| Prop. Lesioned LH | | | −.555 | 2.054 | 51 | −.270 | .7882 |
| Time in Sentence Prod/Comp vs. Time in NH Group | | | 2.381 | .348 | 54 | 6.832 | <.0001 |
| Time in Naming vs. Time in NH Group | | | 1.308 | .318 | 54 | 4.115 | .0001 |
| Time in Spelling vs. Time in NH Group | | | 2.603 | .341 | 54 | 7.640 | <.0001 |
| **Post-hoc analyses** | | | | | | | |
| *Sentence Prod/Comp Treatment Group* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | .720 | .849 | | | | | |
| Residual | .702 | .838 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | −.317 | .501 | 8 | −.633 | .5440 |
| Time (Post- vs. Pre-treatment) | | | 2.520 | .342 | 11 | 7.367 | .0000 |
| Age | | | −.050 | .043 | 8 | −1.179 | .2720 |
| Education | | | .003 | .132 | 8 | .024 | .9810 |
| Prop. Lesioned LH | | | −.656 | 4.090 | 8 | −.160 | .8760 |
| *Naming Treatment Group* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | 1.379 | 1.174 | | | | | |
| Residual | .270 | .520 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | .656 | .370 | 13 | 1.770 | .1000 |
| Time (Post- vs. Pre-treatment) | | | 1.447 | .178 | 16 | 8.121 | .0000 |
| Age | | | −.031 | .044 | 13 | −.698 | .4970 |
| Education | | | .028 | .170 | 13 | .167 | .8700 |
| Prop. Lesioned LH | | | −3.510 | 5.014 | 13 | −.700 | .4960 |
| *Spelling Treatment Group* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | .659 | .812 | | | | | |
| Residual | .655 | .810 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | .048 | .340 | 9 | .142 | .8900 |
| Time (Post- vs. Pre-treatment) | | | 2.742 | .318 | 12 | 8.636 | .0000 |
| Age | | | .016 | .031 | 9 | .529 | .6100 |
| Education | | | −.008 | .248 | 9 | −.031 | .9760 |
| Prop. Lesioned LH | | | −4.024 | 4.721 | 9 | −.852 | .4160 |
| *Natural History Group* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | 1.467 | 1.211 | | | | | |
| Residual | .172 | .415 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | −.661 | .321 | 12 | −2.059 | .0619 |
| Time (Post- vs. Pre-treatment) | | | .139 | .147 | 15 | .947 | .3586 |
| Age | | | .028 | .037 | 12 | .747 | .4692 |
| Education | | | −.116 | .156 | 12 | −.745 | .4708 |
| Prop. Lesioned LH | | | 4.401 | 4.320 | 12 | 1.019 | .3285 |

*Note*. LH = left hemisphere; NH = Natural History.

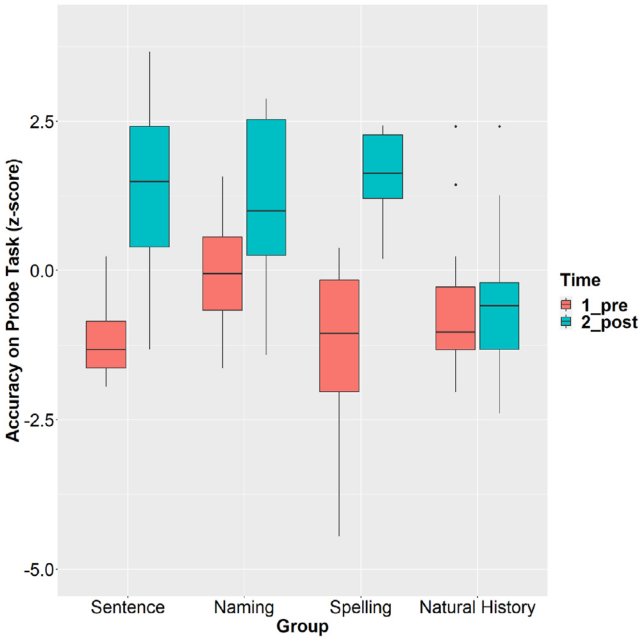


**Fig. 1 — Treatment outcomes. The boxplot shows the median (z-scored) accuracy at pre- (red) and post-testing (blue) for individuals in each of the three treatment groups and in the natural history group. Boxes indicate the range between the 1st and the 3rd quartile of the distribution, and the whiskers mark the minimum and maximum of the distribution.**

### 3.2. *Lesion patterns*

Composite lesion maps for participants in each group are shown in Fig. 2. Although the total proportion of lesioned cortical tissue in the left hemisphere was matched across participant groups, participants in the sentence production/comprehension treatment group exhibited considerable lesion overlap (damage encompassing more than 33% (1/3) of the ROI) in several perisylvian regions (Fig. 2 and Supplementary Table 1) and larger lesions than other groups in the anterior and posterior superior temporal gyrus (aSTG, pSTG), in the posterior and temporo-occipital portions of the middle temporal gyrus (pMTG, tpoMTG), in the posterior portion of the supramarginal gyrus (pSMG) and in the angular gyrus (AG, see Supplementary table 1 for statistics by ROI). Conversely, lesion size in all the parcellations of the IFG was comparable across groups. For all the remaining groups, maps showed the largest overlap in inferior frontal and anterior superior temporal regions; limited overlap was observed in posterior temporal and inferior parietal language areas.

### 3.3. *Functional neuroimaging data*

#### 3.3.1. *Healthy individuals*

3.3.1.1. Accuracy. Participants' performance on the off-line yes/no questions presented at the end of the scanning session was above chance for all participants except two (see Table 3). Mean and standard error for the group of healthy participants are provided in Table 3.

3.3.1.2. Whole-brain analysis. Results of the whole-brain analysis are reported in Fig. 3 and Table 4, reflecting regions that were more active when listening to real stories compared to digitally reversed speech. Overall, healthy participants evinced activation in a large cluster in the left hemisphere, with activation peaking in the aSTG and extending anteriorly and posteriorly to encompass both anterior and posterior temporal, as well as inferior parietal, regions. Within the left hemisphere, a large cluster encompassing most of the IFG, as well as part of the insula and the frontal operculum, was also found. In addition, the left PCG and the left SMA were more active during real stories than during the control condition. The task also yielded activation in the middle and superior frontal gyri (MFG, SFG) and the frontal pole in the left hemisphere. Right-hemisphere activation was considerably less widespread, albeit still significant, with smaller clusters observed in inferior frontal, posterior temporal and inferior parietal regions. Finally, two clusters of activation were found in the left basal ganglia, encompassing the thalamus, the caudate and the putamen, and in the right caudate.

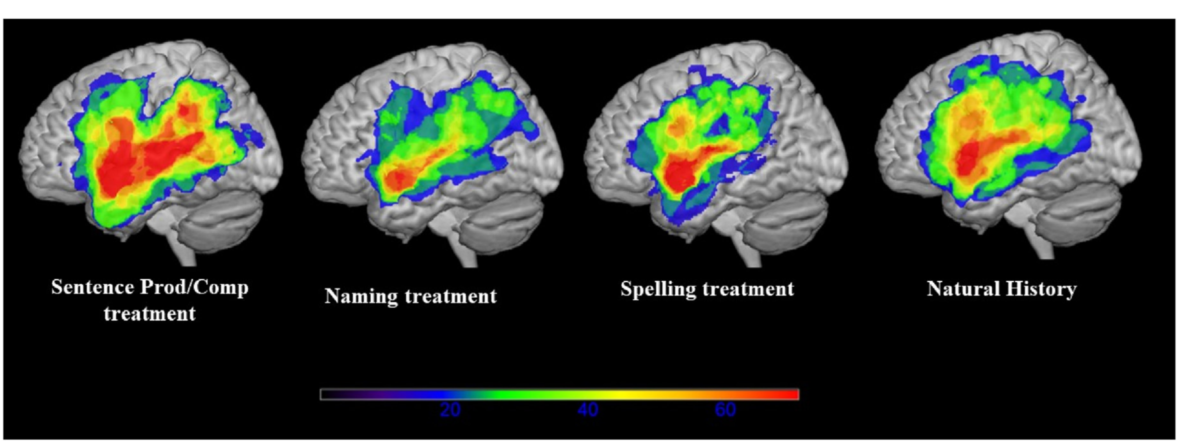


**Fig. 2 — Composite lesion maps displaying the percentage of participants in each group that displayed lesions in each voxel. Warmer colors show regions of greater lesion overlap (min = 1%, max = 70%).**

**Table 3 – Percentage correct responses on the off-line yes/no comprehension questions administered at the end of the scan session for each of the participant groups at each time point. Mean and Standard Error (SE) are provided, together with the number of participants performing above chance at each time point. Asterisks indicate statistical significance for the comparison between each patient group and the group of healthy individuals. *** = $p < .001$ (after applying FDR correction).**

| | Pre-treatment | | | Post-treatment | | |
|---|---|---|---|---|---|---|
| | Mean | SE | # of participants with above-chance performance | Mean | SE | # of participants with above-chance performance |
| Healthy Participants | 84.09 | 3.15 | 20/22 | n/a | | |
| Sentence Treatment | 64.06*** | 5.17 | 6/12 | 62.50*** | 3.77 | 4/12 |
| Naming Treatment | 57.81*** | 5.51 | 3/17 | 63.28*** | 1.42 | 2/17 |
| Spelling Treatment | 62.50*** | 3.95 | 4/13 | 64.38*** | 3.61 | 4/13 |
| Natural History | 56.82*** | 4.17 | 4/16 | 63.13*** | 4.32 | 5/16 |

3.3.2. *Individuals with aphasia*

3.3.2.1. Accuracy. For all participant groups, at all time points, accuracy on the comprehension questions administered at the end of the scanning session was significantly lower than that of healthy participants (Table 3), and several participants showed chance-level performance at both pre- and post-treatment. No significant differences were found between pre- and post-testing accuracy in any of the groups (all *ps* > .1). Notably, due to experimenter errors, data were not available for 15 of the 58 participants with aphasia (Naming treatment: $N = 7$; Spelling Treatment: $N = 3$; Natural History: $N = 5$).

3.3.2.2. Voxel-based analyses. One sample *t*-tests were run within each of the patient groups, at each time point, to find voxels—within regions that were active (or homologous to active regions) in healthy participants—that were significantly active for the *Story* > *Control* contrast at pre- and post-treatment. To do so, regions of interest (ROIs) for all the cortical regions indicated in Table 4 were extracted from the Harvard–Oxford atlas and aggregated to form an external mask that was used to constrain the voxel-based analysis. Results of these analyses are reported in Fig. 4 and Table 5.

Fig. 4 and Table 5 show that, while no suprathreshold activation was found at pre-treatment for any of the patient groups, post-treatment activation was significantly greater than zero in all the three treatment groups, but not in the natural history group. Among the treatment groups, more widespread post-treatment activation was found in the sentence production/comprehension treatment group, compared to the naming and spelling groups. Namely, in the sentence production/comprehension treatment group, post-treatment activation was observed in two right temporal clusters, one more anterior, encompassing the aMTG and aSTG, and extending to include portions of the pMTG and pSTG, and one more posterior, centered on the pMTG and encompassing the temporo-occipital portion of the MTG and pSTG. An additional cluster, encompassing the right IFGtri and IFGoper, and a small portion of the MFG, was also found. Finally, activation

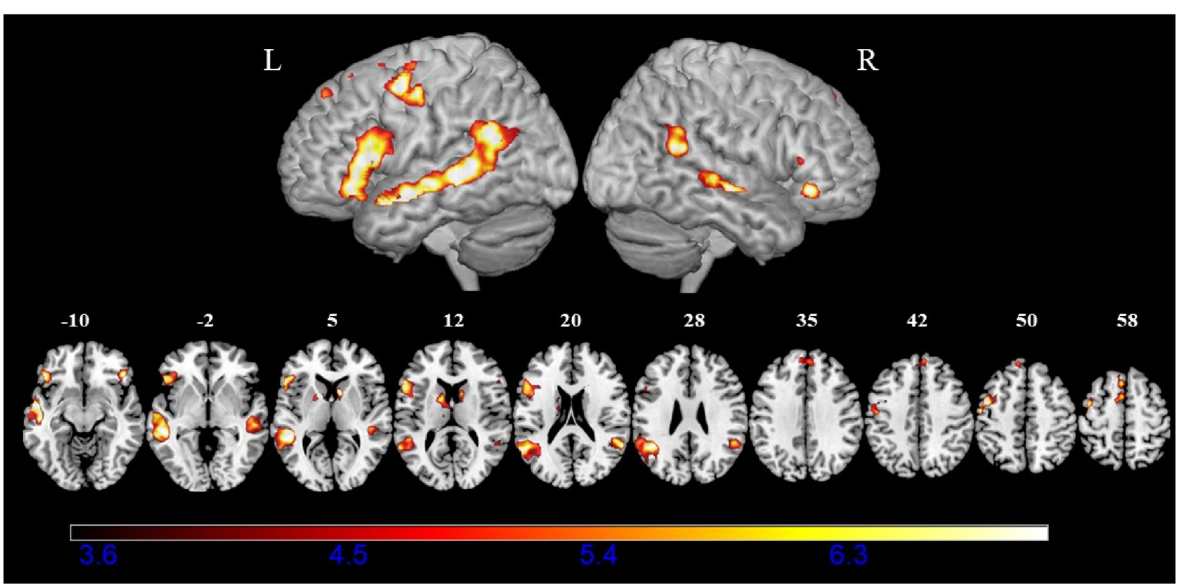


**Fig. 3 – Results of the whole-brain analysis (one-sample *t*-test) conducted on 22 healthy participants ($p < .001$ voxel-level, FWE ($p < .05$) cluster correction, $k \geq 210$ (cluster size = 1680 mm$^3$)). Slices are labeled with z coordinates that mark the exact location on the axial view. The color bar indicates the strength of activation at each voxel, expressed in T-values (range: 3.5 to 7).**

**Table 4 – Clusters of suprathreshold activation ($p < .001$ voxel-level, FWE ($p < .05$) cluster correction, $k > 210$) for *Auditory Stories* > *Control* for healthy individuals ($N = 22$). Region labels are derived from the Harvard–Oxford atlas (Desikan et al., 2006).**

| Cluster FWE-corrected *p* | Cluster Size | Peak T-value | Peak Coordinates | | | Peak Region | Extent | L/R |
|---|---|---|---|---|---|---|---|---|
| | | | x | y | z | | | |
| .0000 | 2882 | 10.161 | −52 | −4 | −12 | aSTG | aMTG, pMTG, tpoMTG, pSTG, Temporal Pole, AG, pSMG, sLOC | L |
| .0000 | 1239 | 8.877 | −56 | 16 | 8 | IFGoper | IFGtri, Frontal Orbital Cortex, Insula, Frontal Operculum | L |
| .0000 | 630 | 9.034 | 52 | −18 | −6 | pSTG | pMTG, tpoMTG | R |
| .0000 | 494 | 9.631 | −46 | 2 | 52 | PCG | MFG, Postcentral | L |
| .0000 | 470 | 6.710 | −12 | 4 | 12 | Caudate | Putamen, Thalamus | L |
| .0000 | 467 | 7.933 | 60 | −50 | 18 | AG | tpoMTG, pSMG | R |
| .0000 | 417 | 8.931 | −4 | 6 | 62 | SMA | SFG | L |
| .0000 | 387 | 6.607 | 6 | 48 | 40 | SFG (R) | SFG (L), Frontal Pole (L) | L/R |
| .0000 | 284 | 9.553 | 8 | 12 | 4 | Caudate | | R |
| .0000 | 210 | 7.890 | 44 | 32 | −8 | white matter | IFGtri, IFGoper, Frontal Orbital Cortex | R |

*Note.* AG = Angular Gyrus; IFGoper = Inferior Frontal Gyrus, pars opercularis; IFGtri = Inferior Frontal Gyrus, pars triangularis; sLOC = Lateral Occipital Cortex, superior division; MFG = Middle Frontal Gyrus; aMTG = Middle Temporal Gyrus, anterior division; pMTG = Middle Temporal Gyrus, posterior division; tpoMTG = Middle Temporal Gyrus, temporo-occipital part; PCG = Precentral Gyrus; SFG = Superior Frontal Gyrus; SMA = Supplementary Motor Area; pSMG = Supramarginal Gyrus, posterior division; aSTG = Superior Temporal Gyrus, anterior division; pSTG = Superior Temporal Gyrus, posterior division.

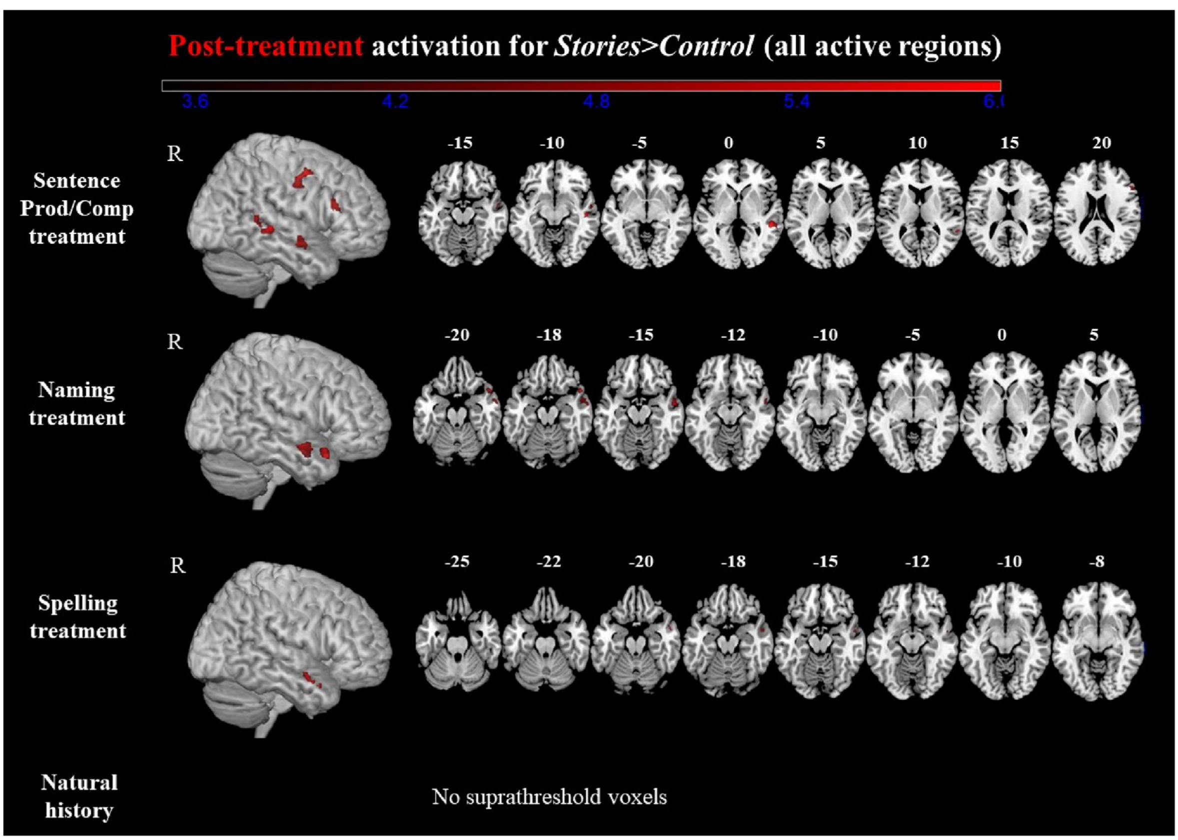


**Fig. 4 – Results of the one-sample *t*-tests conducted on each patient group at post-treatment. T-maps show active voxels at a voxel-level threshold of $p < .001$, after applying a cluster-level FWE correction ($p < .05$), for the contrast *Stories* > *Control*. No voxels survived FWE correction for any of the patient groups at pre-treatment. No significant voxels were found in the natural history group at either pre- or post-treatment. Only right-hemisphere rendered brains are shown, as no significant activation was found in the left hemisphere, for any of the patient groups, at any time point. Slices are labeled with z coordinates that mark the exact location on the axial view. The color bar indicates the strength of activation at each voxel, expressed in T-values (range: 3.5 to 6).**

**Table 5 – Clusters of suprathreshold activation ($p < .001$ voxel-level, FWE ($p < .05$) cluster correction) for *Stories* > *Control* for each of the four patient groups at each time point. Region labels are derived from the Harvard–Oxford atlas (Desikan et al., 2006).**

| Time point | Cluster FWE-corrected $p$ | Cluster Size | Peak T-value | Peak Coordinates | | | Peak Region | Extent | L/R |
|---|---|---|---|---|---|---|---|---|---|
| | | | | x | y | z | | | |
| *Sentence Prod/Comp treatment ($k \geq 60$)* | | | | | | | | | |
| Pre-treatment | No suprathreshold voxels | | | | | | | | |
| Post-treatment | .000 | 185 | 8.335 | 48 | −36 | 0 | pMTG | tpoMTG, pSTG | R |
| | .000 | 131 | 5.972 | 50 | 0 | 48 | PCG | MFG | R |
| | .001 | 126 | 7.002 | 48 | −18 | −10 | pMTG | aMTG, aSTG, pSTG | R |
| | .034 | 60 | 5.462 | 56 | 28 | 20 | IFGtri | IFGoper, MFG | R |
| *Naming treatment ($k \geq 87$)* | | | | | | | | | |
| Pre-treatment | No suprathreshold voxels | | | | | | | | |
| Post-treatment | .0005 | 170 | 5.590 | 64 | −4 | −18 | aMTG | aSTG, pMTG | R |
| | .0195 | 87 | 5.912 | 54 | 16 | −18 | Temporal Pole | | R |
| *Spelling treatment ($k \geq 57$)* | | | | | | | | | |
| Pre-treatment | No suprathreshold voxels | | | | | | | | |
| Post-treatment | .0293 | 57 | 6.066 | 52 | 4 | −20 | aSTG | Temporal Pole | R |
| *Natural History* | | | | | | | | | |
| Pre-treatment | No suprathreshold voxels | | | | | | | | |
| Post-treatment | | | | | | | | | |

*Note*. AG = Angular Gyrus; IFGoper = Inferior Frontal Gyrus, pars opercularis; IFGtri = Inferior Frontal Gyrus, pars triangularis; MFG = Middle Frontal Gyrus; aMTG = Middle Temporal Gyrus, anterior division; pMTG = Middle Temporal Gyrus, posterior division; tpoMTG = Middle Temporal Gyrus, temporo-occipital portion; PCG = Precentral Gyrus; PoCG = Postcentral Gyrus; aSTG = Superior Temporal Gyrus, anterior division; pSTG = Superior Temporal Gyrus, posterior division.

was also found in a cluster peaking in the right PCG. In both the naming and spelling treatment groups, post-treatment activation was limited to right temporal regions, including the aSTG and temporal pole (for both groups), and the aMTG and pMTG (for the naming treatment group only).

3.3.2.3. ROI analyses. The purpose of these analyses was twofold: 1) to compare activation in selected ROIs at pre- and post-treatment across groups and within each treatment group, by accounting for participant-related and ROI-related variability, as well as for age, education, and lesion size, and 2) evaluate the relationship between changes in activation across and within ROIs and changes in performance on measures of word and sentence comprehension. In line with the voxel-based analyses reported in 3.3.2.2., ROI analyses were run within regions that were active in healthy participants, as well as in their right-hemisphere homologues. ROIs were classified as either being part of the **language network** (or **language network homologues**) (i.e., the IFGtri and IFGoper, the frontal orbital cortex, the insula, the frontal operculum, all three MTG parcellations, the aSTG and pSTG, the AG, the pSMG, and the temporal pole), or as belonging to a **domain-general network** (i.e., the Frontal pole, the MFG and SFG, the precentral and postcentral gyri, and the supplementary motor cortex (SMA)). This classification was based on evidence that the IFG, MTG, STG, AG, pSMG and temporal pole in the left hemisphere support lexical, semantic and thematic–syntactic processes that are engaged during several language tasks, including auditory comprehension (see Price, 2012, for a review; also see Thompson & Meltzer-Asscher, 2014), and that activation in left anterior insula and frontal operculum is also associated with sentence processing (Moro et al., 2001; Friederici, Meyer, & Von Cramon, 2000). As for the domain-general network, the right PCG and MFG/SFG are part of the Dorsal Attention Network (Corbetta & Shulman, 2002), which supports visual attention processes, and the right MFG and SFG are also part of the fronto-parietal executive network (Vincent, Kahn, Snyder, Raichle, & Buckner, 2008), which supports executive functions and working memory and also includes the Frontal Pole, bilaterally; finally, the SMA is considered to be part of the cingulo-opercular network, which has been implicated in task control (Geranmayeh et al., 2017; Sestieri, Corbetta, Spadone, Romani, & Shulman, 2014).

3.3.2.3.1. Language network homologues. 3.3.2.3.1.1. Comparison of pre- and post-treatment activation. The full model for the left-hemisphere ROIs revealed no significant effect of *Time* ($t = 1.446$, $p = .148$) and no significant *Group*Time* interaction ($\chi^2 = 5.912$, $p = .116$). The full model for the right-hemisphere ROIs (Table 6) showed a significant effect of *Time* across groups, with activation for the *Story* > *Control* contrast being significantly greater at post- than at pre-treatment. A significant *Group*Time* interaction was also found ($\chi^2 = 9.928$, $p = .019$). Separate analyses conducted within each group (Table 6) revealed a significant increase in activation from pre- to post-treatment (i.e., upregulation) only in the sentence production/comprehension and the spelling treatment groups, and no changes between time points for the naming treatment group or for the natural history (no treatment) group. No effect of age, education or lesion size was found either across or within groups.

3.3.2.3.1.2. Relation between changes in activation and changes in language performance. These analyses were run on the sentence production/comprehension and spelling treatment groups only, as no evidence of upregulation was found in the naming treatment (or the natural history) group in the previous analysis (3.3.2.3.1.1). In addition, analyses were run only within ROIs that showed significantly greater activation for the *Story* > *Control* contrast at post-treatment, to ensure

**Table 6 – Results of the mixed-effects regression analyses conducted on the four groups within the language network homologues. Results are provided for the full model comparing pre- and post-treatment activation across groups, as well as for post-hoc analyses conducted within each group. Variance and Standard Deviation (SD) are displayed for random effects, and beta estimates, Standard Error (SE), t-statistics and p-values are reported for all fixed effects. For the post-hoc analyses, reported p-values reflect FDR (Benjamini & Hochberg, 1995) correction for multiple comparisons.**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Full Model** | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | .027 | .165 | | | | | |
| ROI | .028 | .167 | | | | | |
| Residual | .085 | .291 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | .063 | .052 | 17.5 | 1.212 | .241 |
| Group: N vs. NH | | | .148 | .062 | 51.6 | 2.376 | **.021** |
| Group: Sp vs. NH | | | .099 | .065 | 51.9 | 1.512 | .137 |
| Group: S vs. NH | | | .151 | .071 | 51.6 | 2.130 | **.038** |
| Time (Post- vs. Pre-treatment) | | | .045 | .015 | 1429.0 | 2.979 | **.003** |
| Age | | | −.003 | .002 | 53.0 | −1.180 | .243 |
| Education | | | −.001 | .012 | 51.9 | −.112 | .911 |
| Prop. Lesioned LH | | | −.054 | .038 | 1457.0 | −1.430 | .153 |
| Time in N vs. Time in NH | | | −.031 | .040 | 1429.0 | −.782 | .435 |
| Time in Sp vs. Time in NH | | | .072 | .043 | 1429.0 | 1.683 | .093 |
| Time in S vs. Time in NH | | | .081 | .044 | 1429.0 | 1.859 | .063 |
| **Post-hoc analyses** | | | | | | | |
| *Natural History Group* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | .020 | .142 | | | | | |
| ROI | .028 | .167 | | | | | |
| Residual | .113 | .335 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | −.046 | .061 | 20.2 | −.761 | .456 |
| Time (Post- vs. Pre-treatment) | | | .015 | .033 | 386.0 | .447 | .655 |
| Age | | | .000 | .004 | 14.3 | .097 | .924 |
| Education | | | .026 | .019 | 13.2 | 1.396 | .667 |
| Prop. Lesioned LH | | | −.082 | .077 | 365.1 | −1.056 | .521 |
| *Sentence Prod/Comp Group* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | .027 | .166 | | | | | |
| ROI | .031 | .176 | | | | | |
| Residual | .062 | .248 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | .118 | .095 | 14.0 | 1.240 | .456 |
| Time (Post- vs. Pre-treatment) | | | .096 | .028 | 285.9 | 3.402 | **.003** |
| Age | | | −.001 | .008 | 8.9 | −.195 | .924 |
| Education | | | .003 | .023 | 8.9 | .144 | .889 |
| Prop. Lesioned LH | | | .006 | .068 | 303.8 | .088 | .930 |
| *Naming Group* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | .027 | .165 | | | | | |
| ROI | .019 | .138 | | | | | |
| Residual | .069 | .264 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | .117 | .064 | 23.2 | 1.836 | .317 |
| Time (Post- vs. Pre-treatment) | | | −.016 | .025 | 410.8 | −.653 | .655 |
| Age | | | −.010 | .005 | 14.3 | −2.085 | .222 |
| Education | | | −.019 | .023 | 13.8 | −.859 | .667 |
| Prop. Lesioned LH | | | −.109 | .063 | 431.6 | −1.741 | .330 |
| *Spelling Group* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | .036 | .190 | | | | | |
| ROI | .047 | .218 | | | | | |
| Residual | .088 | .297 | | | | | |
| Fixed Effects | | | Estimate | SE | df | T | p-value |
| (Intercept) | | | .079 | .085 | 20.0 | .922 | .456 |
| Time (Post- vs. Pre-treatment) | | | .087 | .032 | 306.4 | 2.670 | **.016** |

**Table 6 – (*continued*)**

| | | | | | |
|---|---|---|---|---|---|
| Age | −.001 | .006 | 9.9 | −.235 | .924 |
| Education | −.031 | .044 | 10.1 | −.699 | .667 |
| Prop. Lesioned LH | .093 | .108 | 303.7 | .859 | .521 |

*Note*. LH = left hemisphere; N = naming treatment group; NH = natural history group; S = sentence production/comprehension treatment group; Sp = spelling treatment group.

that the relationship between upregulation of activation and changes in language scores was investigated in ROIs for which upregulation of activation resulted in neural response patterns during naturalistic language comprehension that were similar to those of healthy individuals (i.e., greater for real stories than for reversed speech). Thus, the RH IFGtri and IFGoper, the MTG (all three parcellations) and the STG (anterior and posterior division) were included in the analyses on the sentence production/comprehension treatment group, whereas only the RH aSTG and the temporal pole were included in the analyses on the spelling treatment group. Separate analyses were run on each of the two groups, and for each of the language measures (i.e., NNB noun comprehension, NNB verb comprehension, NAVS SCT).

Analyses run within the sentence production/comprehension treatment group showed a significant (or marginally significant, in the case of sentence comprehension, following correction for multiple comparisons) positive association between changes in activation in the language network homologue and improved accuracy on measures of verb and sentence comprehension (Fig. 5). No associations were found in the spelling treatment group (Table 7).

3.3.2.3.2. Domain-general network. 3.3.2.3.2.1. Comparison of pre- and post-treatment activation. The full model for the LH ROIs showed no significant effect of *Time* across groups (Table 8), but a significant *Group**Time* interaction was found ($\chi^2 = 10.966$, $p = .012$), which pointed to significant differences in the effect of *Time* between groups. Post-hoc analyses run within each participant group showed significant changes in activation, in the direction of greater post-vs. pre-treatment activation, only in the spelling treatment group. No changes in activation were found in the sentence production/comprehension or the naming treatment groups, or the natural history group. No effect of age, education or lesion size were found across or within groups. Results of the analyses conducted on the RH ROIs (Table 8) were consistent with results obtained from the LH ROIs, in that they revealed a significant *Group**Time* interaction ($\chi^2 = 10.387$, $p = .015$), with separate analyses within each group showing significant increase in activation from pre-to post-treatment in the spelling treatment group, but not in any of the other groups.

3.3.2.3.2.2. Relation between changes in activation and changes in language performance. Because none of the participant groups showed significant upregulation of activation on the ROI analyses and evidence of post-treatment suprathreshold activation on the voxel-based analysis, analyses on the relationship between changes in activation in the domain-general network and changes in word and sentence comprehension were not run for any of the participant groups.

## 4. Discussion

This study investigated the effect of treatment provided for deficits in different language domains on the neural activation patterns of individuals with chronic aphasia. Four groups of participants (three assigned to receive treatment for either sentence production/comprehension, naming or spelling, deficits, and a natural history group that did not receive treatment) completed behavioral testing and an fMRI story comprehension task at baseline (pre-treatment) and three months later (post-treatment), coinciding with the cessation

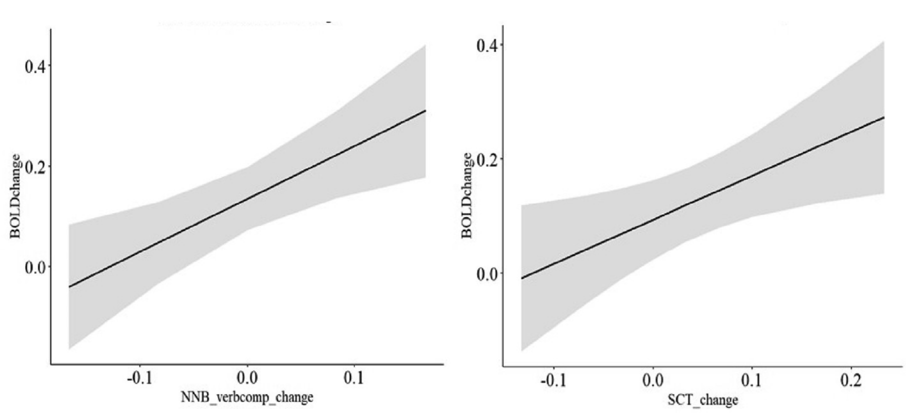


**Fig. 5 – Plots displaying the relationship between changes in BOLD signal between time points (post-minus pre-treatment) and the change in accuracy on measures of verb comprehension (NNB_verbcomp_change) and sentence comprehension (SCT_change) for the sentence production/comprehension treatment group.**

**Table 7 – Results of the mixed-effects regression analyses investigating the relationship between changes in activation between time points and changes in word and sentence comprehension accuracy within ROIs that were included in the language network homologue and that showed suprathreshold activation on the voxel-based analysis reported in Fig. 4 and Table 5. Variance and Standard Deviation (SD) are reported for the random effect of ROI. Beta estimates, Standard Error (SE), degrees of freedom (df), as well as T-statistics and p-values, are reported for fixed effects. *p*-values reflect FDR correction for multiple comparisons (i.e., separate analyses for each group).**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Sentence Production/Comprehension Treatment group** | | | | | | | |
| *Language Measure = Change on NNB Noun Comprehension* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| ROI | .000 | .000 | | | | | |
| Residual | .105 | .324 | | | | | |
| Fixed Effects | | | Estimate | SE | df | T | p-value |
| (Intercept) | | | .150 | .036 | 94.0 | 4.130 | .001 |
| change on NNB NC | | | 1.739 | 1.196 | 94.0 | 1.454 | .224 |
| *Language Measure = Change on NNB Verb Comprehension* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| ROI | .000 | .000 | | | | | |
| Residual | .098 | .312 | | | | | |
| Fixed Effects | | | Estimate | SE | df | T | p-value |
| (Intercept) | | | .121 | .032 | 94.0 | 3.798 | .001 |
| change on NNB VC | | | 1.051 | .343 | 94.0 | 3.063 | .017 |
| *Language Measure = Change on NAVS Sentence Comprehension* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| ROI | .000 | .000 | | | | | |
| Residual | .101 | .318 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | .104 | .034 | 94.0 | 3.072 | .006 |
| change on NAVS SCT | | | .768 | .318 | 94.0 | 2.418 | .053 |
| **Spelling Treatment group** | | | | | | | |
| *Language Measure = Change on NNB Noun Comprehension* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| ROI | .000 | .000 | | | | | |
| Residual | .165 | .407 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | .031 | .079 | 25.0 | .391 | .728 |
| change on NNB NC | | | 1.657 | 1.576 | 25.0 | 1.051 | .364 |
| *Language Measure = Change on NNB Verb Comprehension* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| ROI | .000 | .000 | | | | | |
| Residual | .159 | .398 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | .059 | .077 | 25.0 | .763 | .679 |
| change on NNB VC | | | .690 | .458 | 25.0 | 1.505 | .223 |
| *Language Measure = Change on NAVS Sentence Comprehension* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| ROI | .000 | .000 | | | | | |
| Residual | .171 | .414 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | .030 | .084 | 25.0 | .352 | .728 |
| change on NAVS SCT | | | −.598 | 1.222 | 25.0 | −.490 | .629 |

*Note.* NAVS = Northwestern Assessment of Verbs and Sentences; NC = Noun Comprehension; NNB = Northwestern Naming Battery; SCT = Sentence Comprehension Test; VC = Verb Comprehension.

of a 12-week treatment period provided for all treatment group participants (with the same amount and intensity across groups). First, whole-brain voxel-based analyses were run on the fMRI story comprehension task within a group of healthy individuals, to detect active voxels for the *Story* > *Control* contrast (i.e., those significantly greater than zero). Similar analyses, albeit restricted to regions that were active in healthy participants or right hemisphere homologous to active regions, were run on each patient group at each time point (pre-treatment, post-treatment). Region-Of-Interest (ROI)-based analyses were then run within the same set of regions (i.e., ROIs that were active in healthy participants, as well as their right-hemisphere homologues) to evaluate changes in activation (post- vs. pre-treatment) across groups and within each treatment group. Lastly, in groups and ROIs for which significant changes in activation were detected on the ROI-based analysis and suprathreshold post-treatment activation for the *Story* > *Control* contrast was detected on

**Table 8 – Results of the mixed-effects regression analyses conducted on the four participant groups within the left and right ROIs that were part of the domain-general network. Results are provided for the full model comparing pre- and post-treatment activation across groups, as well as for post-hoc analyses conducted within each group. Variance and Standard Deviation (SD) are displayed for random effects. Beta estimates, Standard Error (SE), degrees of freedom (df), as well as t-statistics and p-values are reported for all fixed effects.**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Full Model** | | | | | | | |
| **Left Hemisphere** | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | .014 | .117 | | | | | |
| ROI | .003 | .056 | | | | | |
| Residual | .038 | .194 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | −.045 | .027 | 12.8 | −1.657 | .122 |
| Group: N vs. NH | | | .064 | .046 | 52.0 | 1.391 | .170 |
| Group: Sp vs. NH | | | .008 | .048 | 51.8 | .159 | .874 |
| Group: S vs. NH | | | .069 | .052 | 52.1 | 1.322 | .192 |
| Time (Post- vs. Pre-treatment) | | | −.001 | .014 | 743.0 | −.099 | .921 |
| Age | | | .000 | .002 | 52.1 | −.004 | .997 |
| Education | | | −.001 | .009 | 51.8 | −.064 | .949 |
| Prop. Lesioned LH | | | −.011 | .063 | 748.3 | −.172 | .864 |
| Time in N vs. Time in NH | | | .029 | .036 | 743.0 | .795 | .427 |
| Time in Sp vs. Time in NH | | | .107 | .039 | 743.0 | 2.777 | **.006** |
| Time in S vs. Time in NH | | | −.016 | .040 | 743.0 | −.406 | .685 |
| **Post-hoc analyses** | | | | | | | |
| *Natural History Group* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | .017 | .129 | | | | | |
| ROI | .003 | .055 | | | | | |
| Residual | .033 | .181 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | −.082 | .041 | 15.8 | −2.005 | .248 |
| Time (Post- vs. Pre-treatment) | | | −.031 | .024 | 200.0 | −1.294 | .263 |
| Age | | | .000 | .003 | 12.8 | .091 | .929 |
| Education | | | −.012 | .017 | 12.6 | −.717 | .947 |
| Prop. Lesioned LH | | | .124 | .100 | 152.7 | 1.246 | .359 |
| *Sentence Prod/Comp Group* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | .003 | .051 | | | | | |
| ROI | .005 | .069 | | | | | |
| Residual | .029 | .170 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | −.029 | .041 | 12.0 | −.709 | .524 |
| Time (Post- vs. Pre-treatment) | | | −.047 | .026 | 148.4 | −1.807 | .146 |
| Age | | | −.002 | .003 | 9.0 | −.633 | .724 |
| Education | | | .001 | .009 | 9.1 | .074 | .947 |
| Prop. Lesioned LH | | | −.290 | .137 | 133.1 | −2.118 | .144 |
| *Naming Group* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | .016 | .126 | | | | | |
| ROI | .001 | .036 | | | | | |
| Residual | .043 | .206 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | .028 | .043 | 15.2 | .653 | .524 |
| Time (Post- vs. Pre-treatment) | | | −.003 | .027 | 212.9 | −.100 | .921 |
| Age | | | −.007 | .004 | 13.9 | −1.779 | .298 |
| Education | | | .024 | .018 | 13.6 | 1.326 | .828 |
| Prop. Lesioned LH | | | −.145 | .131 | 191.7 | −1.108 | .359 |
| *Spelling Group* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | .014 | .118 | | | | | |
| ROI | .012 | .110 | | | | | |
| Residual | .037 | .1924 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | −.084 | .057 | 12.2 | −1.477 | .330 |

(continued on next page)

**Table 8 – (continued)**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Time (Post- vs. Pre-treatment) | | | .076 | .029 | 161.0 | 2.666 | **.032** |
| Age | | | .006 | .004 | 9.9 | 1.566 | .298 |
| Education | | | .002 | .029 | 10.1 | .069 | .947 |
| Prop. Lesioned LH | | | −.001 | .133 | 175.6 | −.011 | .991 |
| **Full Model** | | | | | | | |
| **Right Hemisphere** | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | .024 | .156 | | | | | |
| ROI | .005 | .069 | | | | | |
| Residual | .038 | .195 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | −.087 | .034 | 14.4 | −2.562 | **.022** |
| Group: N vs. NH | | | .113 | .059 | 52.0 | 1.922 | .060 |
| Group: Sp vs. NH | | | .009 | .061 | 51.9 | .144 | .886 |
| Group: S vs. NH | | | .112 | .067 | 52.1 | 1.674 | .100 |
| Time (Post- vs. Pre-treatment) | | | .026 | .014 | 743.0 | 1.897 | .058 |
| Age | | | −.002 | .002 | 52.1 | −.862 | .393 |
| Education | | | −.009 | .011 | 51.9 | −.782 | .438 |
| Prop. Lesioned LH | | | .069 | .065 | 779.4 | 1.057 | .291 |
| Time in N vs. Time in NH | | | .034 | .036 | 743.0 | .927 | .354 |
| Time in Sp vs. Time in NH | | | .122 | .039 | 743.0 | 3.143 | **.002** |
| Time in S vs. Time in NH | | | .038 | .040 | 743.0 | .960 | .338 |
| **Post-hoc analyses** | | | | | | | |
| *Natural History Group* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | .019 | .138 | | | | | |
| ROI | .006 | .075 | | | | | |
| Residual | .043 | .209 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | −.146 | .047 | 15.4 | −3.083 | **.028** |
| Time (Post- vs. Pre-treatment) | | | −.022 | .028 | 200.2 | −.802 | .637 |
| Age | | | −.002 | .003 | 13.1 | −.686 | .926 |
| Education | | | −.006 | .018 | 12.9 | −.346 | .841 |
| Prop. Lesioned LH | | | .064 | .116 | 177.4 | .548 | .584 |
| *Sentence Prod/Comp Group* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | .005 | .068 | | | | | |
| ROI | .010 | .098 | | | | | |
| Residual | .027 | .163 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | −.024 | .052 | 12.6 | −.462 | .869 |
| Time (Post- vs. Pre-treatment) | | | .016 | .025 | 148.3 | .632 | .637 |
| Age | | | .000 | .003 | 8.9 | −.096 | .926 |
| Education | | | −.002 | .010 | 9.0 | −.240 | .841 |
| Prop. Lesioned LH | | | −.085 | .137 | 154.8 | −.618 | .584 |
| *Naming Group* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | .035 | .187 | | | | | |
| ROI | .000 | .004 | | | | | |
| Residual | .034 | .185 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | −.002 | .057 | 13.8 | −.043 | .966 |
| Time (Post- vs. Pre-treatment) | | | .011 | .024 | 213.2 | .473 | .637 |
| Age | | | −.009 | .006 | 14.0 | −1.569 | .556 |
| Education | | | .005 | .025 | 13.9 | .204 | .841 |
| Prop. Lesioned LH | | | −.216 | .114 | 112.4 | −1.896 | .122 |
| *Spelling Group* | | | | | | | |
| Random Effects | Variance | SD | | | | | |
| Participant | .033 | .182 | | | | | |
| ROI | .012 | .112 | | | | | |
| Residual | .039 | .198 | | | | | |
| Fixed Effects | | | Estimate | SE | df | t | p-value |
| (Intercept) | | | −.128 | .071 | 14.6 | −1.801 | .184 |
| Time (Post- vs. Pre-treatment) | | | .100 | .029 | 161.0 | 3.412 | **.004** |

**Table 8 – (continued)**

| | | | | | |
|---|---|---|---|---|---|
| Age | .001 | .006 | 10.0 | .204 | .926 |
| Education | −.047 | .042 | 10.1 | −1.115 | .841 |
| Prop. Lesioned LH | .288 | .140 | 175.9 | 2.065 | .122 |

*Note*. LH = left hemisphere; N = naming treatment group; NH = natural history group; S = sentence production/comprehension treatment group; Sp = spelling treatment group.

the voxel-based analysis, the relationship between activation changes and improvements in word and sentence comprehension accuracy was evaluated.

### 4.1. *Treatment efficacy (behavior)*

As expected, and as reported in previous studies, behavioral results showed that treatment for deficits in all language domains was effective, resulting in increased language ability (see Barbieri et al., 2019, for sentence production/comprehension treatment; Gilmore et al., 2019 and Johnson et al., 2019, for naming treatment, and Purcell et al., 2019, for spelling treatment). Albeit targeting different language functions, all three types of treatment employed in this study adopted an “impairment-based approach”, i.e., treatment was focused on a specific aspect of language processing that was impaired in the clinical population for which treatment was designed. In line with previous research using similar approaches to language recovery (e.g., Conroy, Sage, & Lambon Ralph, 2009; Fridriksson, Morrow-Odom, Moser, Fridriksson, & Baylis, 2006; Kiran & Thompson, 2003; Schwartz, Saffran, Fink, Myers, & Martin, 1994; Thompson et al., 2010; Wierenga et al., 2006), and with the results of a recent Phase III randomized clinical trial (Breitenstein et al., 2017), these results show that language interventions grounded in psycholinguistic models of language processing effectively improve trained language functions, and that, conversely, such improvement does not occur simply through repeated exposure to language assessments (i.e., as seen in the natural history group).

### 4.2. *Accuracy on fMRI story comprehension task*

Healthy participants' accuracy on the offline yes/no comprehension questions was high, albeit not at ceiling, and two healthy individuals (out of 22) exhibited chance-level performance. We note that accuracy on these questions is not an accurate reflection of participants’ language comprehension skills, as questions were administered 30–40 min after completing the fMRI task. In addition to word/sentence comprehension, successful performance on this task relies on the ability to encode verbal information, and to store it and retrieve it from long-term memory. Chance-level accuracy in these two healthy participants may have therefore stemmed from either failure to encode some of the information in the stories, or from decay of information over time. We note that the latter is not to be considered a reason for excluding these participants from the analyses, as memory tests typically assess delayed recall of information after a delay (maximum of 30-min).

As for individuals with aphasia, performance was at chance in more than half of the participants both prior to and following treatment, and no group-level significant changes in accuracy were observed between time points. We note that lack of improvement in accuracy on the comprehension questions does not undermine our conclusions about treatment efficacy. Indeed, we did not expect improvement on the offline questionnaire since the comprehension questions were included to promote attention to the task during scanning sessions and should not be considered a reliable measure of participants’ narrative comprehension for the reasons stated above.

### 4.3. *Neural plasticity*

Whole-brain analyses revealed that, when listening to short narratives (compared to reversed speech), healthy participants activated a large fronto-temporal-parietal network in the left hemisphere, as well as a subset of homologous regions in the right hemisphere. These results are in line with previous studies using similar tasks (e.g. Crinion & Price, 2005; Skipper et al., 2005; Wilson et al., 2008; Yarkoni et al., 2008), and provide support to the use of naturalistic narrative comprehension tasks, in which task demands are kept to a minimum, as a way to elicit robust activation within the language network (see also Wilson, Bautista, Yen, Lauderdale, & Eriksson, 2017).

In individuals with aphasia, at the pre-treatment time point, none of the patient groups showed significant (i.e., greater than zero) activation for *Stories* > *Control* in either left or right-hemisphere regions that were active in healthy participants or that were homologous to those regions. The absence of left-hemisphere activation on the fMRI task is in line with a previous study by Crinion and Price (2005), who found no left-hemisphere activation for listening to short stories (over a reversed speech control condition) in participants with lesions affecting the left temporal lobe, against normal-like activation patterns in a group of individuals with aphasia and lesions not affecting the left temporal lobe. As noted by Crinion and Price (2005), the lack of activation in the left hemisphere of individuals with aphasia could be due to reduced sensitivity to BOLD signal detection in the damaged cortex. Alternatively, extensive damage of the left middle/anterior temporal regions that support lexical-semantic processing, a characteristic of all our patient groups, may have been sufficient to disrupt connectivity within the entire language network, thereby preventing activation of the language network during the story comprehension fMRI task (see also Sims et al., 2016).

At post-treatment, activation was observed only within right-hemisphere regions and only within the three treatment groups, whereas the natural history group showed no significant activation at post-treatment. Notably, all treatment groups exhibited post-treatment activation of the right anterior temporal cortex (including the anterior portion of the middle and superior temporal gyri, and the temporal pole). Although activation of these regions in healthy participants did not survive correction for multiple comparisons in the present study, research using similar paradigms has reported right anterior temporal lobe activation in healthy participants (Crinion & Price, 2005; Wilson et al., 2017). Therefore, these results suggest that individuals who underwent language treatment, independent of the language function that was targeted during treatment, showed some normalization of activation patterns during narrative comprehension. At post-treatment, participants in the sentence production/comprehension treatment group also activated—contrary to the naming and spelling treatment groups—right inferior frontal and posterior temporal regions, which were also significantly active in healthy participants on the same task. These findings suggest that individuals who underwent interventions for sentence processing deficits exhibited—albeit only in the right hemisphere—activation patterns that were more similar (compared to the naming or spelling treatment groups) to those evinced by healthy individuals, as further discussed below.

One important caveat is that the conclusions that can be drawn from the results these analyses do not take into consideration the role that baseline differences between treatment groups (i.e., age, education and lesion size) and variability among individuals within each group play in shaping recovery. These effects were evaluated by the ROI-based, mixed-effect regression analyses. When accounting for age, education, the extent of lesions within the language network, and individual differences, analyses run within language network homologues (i.e., inferior frontal gyrus, insulo-opercular region, anterior and posterior middle/superior temporal regions, as well as inferior parietal regions) revealed significant upregulation in both the sentence production/comprehension and the spelling, but not the naming, treatment group. Importantly, although changes in activation were found in the language homologue network in its entirety, only a subset of these regions showed suprathreshold post-treatment activation for the *Story* > *Control* contrast on the voxel-based analysis, suggesting that only in selected ROIs within the network, changes in activation resulted in greater activation for real stories than for reversed speech, i.e., in the emergence of neural response patterns to naturalistic language comprehension that were similar to those of healthy individuals. Follow-up regression analyses conducted within these selected ROIs revealed that upregulation of these regions was positively associated with improved accuracy on measures of verb and sentence comprehension in the sentence production/comprehension, but not in the spelling, treatment group.

Taken together, results of the voxel-based and ROI-based analyses indicate that participants who received treatment for sentence processing (i.e., Treatment of Underlying Forms, TUF, Thompson & Shapiro, 2005) showed clear evidence of neural plasticity, as suggested by post-treatment activation increases within right-hemisphere language network homologues that were active in healthy participants on the same task, and by significant associations between activation changes and improved verb and sentence comprehension. These results are in line with our predictions and demonstrate an unequivocal link between activation of right-hemisphere language network homologues and improved language performance. Because TUF improves access to verbs and their lexical representations, by emphasizing the lexical–semantic properties of verbs (thematic roles, i.e., the roles assigned to the participants in sentences) and thematic-syntactic mapping in sentences, activation increases in language network homologues on the story comprehension task may reflect treatment-induced improvements in lexical, sentence and discourse-level comprehension in this group. Alternatively, treatment for sentence comprehension and production may have helped restore functional activity within right hemisphere regions supporting sentence processing in healthy listeners, which also support comprehension of short narratives (see Yarkoni et al., 2008; see also Crinion & Price, 2005; Wilson et al., 2008), in line with one of the principles for promoting neuroplasticity following brain damage, according to which language interventions that target a specific language function lead to improvement of that function and of the associated networks (i.e., specificity rebuilds neural networks, Kleim & Jones, 2008; Kiran & Thompson, 2019).

Upregulation of right hemisphere homologues of the language network was also observed in the spelling treatment group. This finding was unexpected, both based on our predictions and on the findings of previous studies (De Marco et al., 2018; Purcell et al., 2017, 2019), in which evidence of neural plasticity following spelling treatment was found in left perilesional regions. Importantly, upregulation of activation in this group was not significantly associated with performance on word or sentence comprehension, suggesting that activation changes in this group did not result from improved lexical, sentence, or discourse-level comprehension following language intervention. One possible interpretation of these results is that the spelling treatment, which also emphasized listening and repetition, may have improved phonological processing, thereby resulting in better discrimination of real speech from reversed speech and increased post-treatment BOLD signal for the *Story* > *Control* contrast.

For the naming treatment group, although voxel-based analyses showed activation of right anterior temporal regions at post- (and not pre-) treatment, the lack of significant upregulation on the ROI-based analyses suggests that the findings of the voxel-based analyses may have been either driven by some participants, or accounted for by age, education or lesion size. Results obtained from this group, which were not in line with our predictions, suggest that language interventions targeting lexical-semantics may not yield robust, group-level upregulation of activation on a naturalistic task that engages processes beyond single-word comprehension. This also indicates that the task employed as primary measure of neural plasticity may have a substantial effect on the outcome and should be carefully chosen.

### 4.4. Significance of recruitment of right-hemisphere regions during recovery

Overall, results of this study indicate that language interventions that focus on sentence-level, rather than word-level, production and comprehension have the potential to result in more normal-like activation patterns and greater neural plasticity on naturalistic tasks of language comprehension. Importantly, neural plasticity was only observed in the right hemisphere, in regions that were activated by healthy participants on the same task and that are homologous to the left-hemisphere language network. These findings support previous studies reporting increased activation and/or increased connectivity, following various types of language interventions, in right-hemisphere regions (Barbieri et al., 2019; Breier, Maher, Novak, & Papanicolaou, 2006; Kiran et al., 2015; Musso et al., 1999; Raboyeau et al., 2008; Thompson et al., 2010; Wan, Zheng, Marchina, Norton, & Schlaug, 2014). In one of these studies, conducted on the same agrammatic participants recruited for this study but using a sentence-picture verification task as neuroimaging outcome measure (Barbieri et al., 2019), significant treatment-induced upregulation was observed in right-hemisphere regions that largely overlapped with those found in this study (i.e., the pars triangularis of the inferior frontal gyrus, and the posterior superior and middle temporal gyri), and were positively related to changes in offline and online sentence comprehension. Together, these two studies provide compelling evidence that recruitment of right hemisphere regions is a viable route for recovery of sentence processing. On a more general level, this buttresses recent observations that language recovery often engages pre-existing networks (or their right-hemisphere homologs) that support language processing in healthy individuals (Stockert et al., 2020; Turkeltaub et al., 2011).

### 4.5. Effect of participant-related variables on neural recovery

The results of this study also offer some insights regarding the role of participant-related variables on aphasia recovery patterns. As noted, the individuals assigned to the four groups were matched for months post-onset of stroke and aphasia severity, but differed in age and education. However, differences in activation changes over time between the three treatment groups remained significant even when accounting for age and education, suggesting that differences between groups were not due to differences in demographics.

Similar conclusions can be drawn for lesion size. Although overall lesion size (i.e., across all left-hemisphere cortical areas) was matched across groups, participants in the sentence production/comprehension treatment group showed larger lesions in some of the regions that are part of the language network, such as the posterior middle and superior temporal gyrus, the anterior superior temporal gyrus and the angular gyrus. For this reason, in all ROI-based analyses run within the language network homologue, the proportion of lesioned tissue in the left-hemisphere language network was always included as a covariate. Results showed no associations between the extent of lesions to the language network and activation changes within language network homologues in any of the groups; in addition, differences in activation changes between the three treatment groups remained significant even when accounting for this variable. The lack of effect of lesion size on activation changes is at odds with some studies of aphasia recovery (Hope et al., 2013; Selnes, Knopman, Niccum, Rubens, & Larson, 1983), but in line with others (Skipper-Kallal, Lacey, Xing, & Turkeltaub, 2017), and suggests that other variables (namely, the type of treatment provided and the task used to detect activation changes) may have a greater impact on activation patterns associated with aphasia recovery.

### 4.6. Role of domain-general regions in language recovery

One of the aims of the present study was to investigate the role of domain-general regions in aphasia recovery. In this study, in line with our predictions and with our previous research (Barbieri et al., 2019), we found post- (but not pre-) treatment activation of the right precentral and middle frontal gyri in the sentence production/comprehension treatment group. However, these results were found only in the voxel-based analysis, and were not supported by the ROI-based analysis, in which no significant changes in activation were found in the sentence treatment group. As earlier mentioned, one possible reason for these inconsistency in the results of the two analyses may be due to individual variability, which was accounted for in the ROI (but not the voxel) based analysis. Namely, the findings of the voxel-based analyses may have been either driven by some participants, or accounted for by age, education or lesion size. Conversely, participants in the spelling treatment group showed significant upregulation of activation in both left and right ROIs included in the domain-general network on the ROI-based analysis, but no evidence of suprathreshold activation in any of the domain-general regions on the voxel-based analysis. In summary, evidence of neural plasticity in domain-general regions was questionable in both groups, given the inconsistency in the results of the two analyses.

The lack of robust, group-level upregulation of activation within domain-general regions in the sentence treatment group is at odds with our previous findings on the same participants using a picture verification task (Barbieri et al., 2019), in which upregulation of right-hemisphere ROIs that were part of the Dorsal Attention Network (DAN, Corbetta & Shulman, 2002) was found both on a whole-brain, voxel-level analysis and on an ROI-based analysis, and was also strongly associated with improved offline and online sentence comprehension. However, we note that the picture verification task used in Barbieri et al. (2019) engendered greater task demands than the story comprehension task employed in this study, as it required long-distance dependency resolution in sentences, a process that relies on working memory (King & Kutas, 1995; Ness & Meltzer-Asscher, 2017; Phillips, Kazanina, & Abada, 2005; Wanner & Maratsos, 1978). Conversely, the story comprehension task employed in this study likely engaged attentional resources to a lesser extent in that the stories consisted of simple, active sentences with no long-distance dependencies. Future studies will need to further evaluate the extent to which treatment focused on different domains of language results in recruitment of domain-general resources using both low-demand and high-demand language tasks.

## 5. Limitations of the current study

The current study bears several limitations. First, as discussed above, participants were recruited from different research centers because they met criteria for different language deficits, which resulted in pre-treatment heterogeneities across groups in terms of language profiles. Future studies need to evaluate both the behavioral and neural effects of different treatments on a more homogeneous group of participants. In addition, because participants in the three treatment groups underwent testing at different research centers, structural and functional images were collected using different scanners, and—although harmonization among scanners was carried out prior to undertaking the study—the possibility that minor inconsistencies among scanners may have affected functional activation patterns cannot be excluded.

Limitations of the study also encompass the design of our fMRI task. We note that the control condition used (i.e., reversed speech) was not matched in duration to the active experimental condition (i.e., real stories). In addition, while participants were monitored on-line for adherence to the task (i.e., button-press to indicate if they were listening to a real story or reversed speech), the task did not directly measure performance accuracy. Although participants were asked questions about the stories at the end of each scanning session, due to the interference exerted by other tasks and to the variable (up to 40 min) time interlapsing between the presentations of the stories and the comprehension questions, responses to these questions were not considered a reliable measure of participants' performance. Future studies using a story comprehension task should consider including comprehension questions following each story, rather than at the end of the task.

Future studies will also need to address the question of how and if treatment-induced changes in activation are maintained over time. The results of the present study offered a pre-vs. post-treatment comparison of behavioral accuracy and activation patterns, however the investigation of the long-term effects of language interventions on neural reorganization requires further study.

## 6. Conclusions

This study is the first to compare the effects of language interventions provided for different language domains on patterns of neural recovery in a large group of people with chronic aphasia. The results suggest that the type of language intervention provided influences neural recovery and the task used to evaluate treatment-induced changes in activation influences study outcomes. Namely, when language treatment is provided on identical schedules with equal intensity, sentence-level treatment—by improving verb and sentence comprehension—results in upregulation of activation on a naturalistic story comprehension task, whereas, word-level treatments result in less robust (for spelling) or no (for naming) upregulation. In addition, the study provides support for the claim that recruitment of right hemisphere regions is viable for supporting the recovery of language function, particularly within regions that support language processing in healthy participants.

## CRediT author statement

**Elena Barbieri:** Conceptualization, Methodology, Formal Analysis, Investigation, Writing—Original Draft, Writing—Review & Editing, Visualization.

**Cynthia K. Thompson**: Conceptualization, Resources, Writing—Original Draft, Writing—Review & Editing, Supervision, Project Administration, Funding Acquisition.

**James Higgins:** Methodology, Formal Analysis, Software, Investigation, Writing—Review & Editing.

**David Caplan:** Conceptualization, Resources, Writing—Review & Editing, Project Administration, Funding Acquisition.

**Swathi Kiran:** Conceptualization, Resources, Writing—Review & Editing, Project Administration, Funding Acquisition.

**Brenda Rapp:** Conceptualization, Resources, Writing—Review & Editing, Project Administration, Funding Acquisition.

**Todd Parrish**: Conceptualization, Methodology, Software, Resources, Writing—Review & Editing, Project Administration, Funding Acquisition.

## Funding sources

This study was supported by the NIH-NIDCD Clinical Research Center grant P50DC012283 (PI: Cynthia K. Thompson).

## Open Practices section

The study in this article earned Preregistered badges for transparent practices. This study was registered on ClinicalTrials.gov in August 2013 (ClinicalTrials.gov identifier: NCT01927302; https://clinicaltrials.gov/ct2/show/NCT01927302?cond=Aphasia&cntry=US&state=US%3AMA&rank=7)

## Declaration of competing interest

Drs Thompson and Kiran serve as Associate Editors for Cortex. The other authors have no conflicts of interest to disclose.

## Acknowledgments

We are grateful to Jennifer Mack, Aya Meltzer-Asscher, Jeremy Purcell, and Jim Kloet for help with the fMRI task design, and to Sarah Chandler, Brianne Chiappetta, Eduardo Europa, Jeffrey Johnson, Erin Meier, and Jennifer Shea for assistance with data collection and language training.

## Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cortex.2022.11.008.

REFERENCES


Abel, S., Weiller, C., Huber, W., & Willmes, K. (2014). Neural underpinnings for model-oriented therapy for aphasic word production. *Neuropsychologia, 57*, 154—167.

Allen, L., Mehta, S., Andrew McClure, J., & Teasell, R. (2012). Therapeutic interventions for aphasia initiated more than six months post stroke: A review of the evidence. *Topics in Stroke Rehabilitation, 19*(6), 523—535.

Alpert, K., Kogan, A., Parrish, T., Marcus, D., & Wang, L. (2016). The Northwestern University Neuroimaging Data Archive (NUNDA). *NeuroImage, 124*, 1131—1136.

Ashburner, J. (2007). A fast diffeomorphic image registration algorithm. *NeuroImage, 38*(1), 95—113.

Baayen, R. H., Davidson, D. J., & Bates, D. M. (2008). Mixed-effects modeling with crossed random effects for subjects and items. *Journal of Memory and Language, 59*(4), 390—412.

Barbieri, E., Mack, J., Chiappetta, B., Europa, E., & Thompson, C. K. (2019). Recovery of offline and online sentence processing in aphasia: Language and domain-general network neuroplasticity. *Cortex; a Journal Devoted To the Study of the Nervous System and Behavior, 120*, 394—418.

Bates, D., Mächler, M., Bolker, B., & Walker, S. (2014). *Fitting linear mixed-effects models using lme4*. https://doi.org/10.48550/arXiv.1406.5823.

Behzadi, Y., Restom, K., Liau, J., & Liu, T. T. (2007). A component based noise correction method (CompCor) for BOLD and perfusion based fMRI. *NeuroImage, 37*(1), 90—101.

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological), 57*(1), 289—300.

Binder, J. R., Westbury, C. F., McKiernan, K. A., Possing, E. T., & Medler, D. A. (2005). Distinct brain systems for processing concrete and abstract concepts. *Journal of Cognitive Neuroscience, 17*(6), 905—917.

Bonakdarpour, B., Parrish, T. B., & Thompson, C. K. (2007). Hemodynamic response function in patients with stroke-induced aphasia: Implications for fMRI data analysis. *NeuroImage, 36*(2), 322—331.

Bonilha, L., Gleichgerrcht, E., Nesland, T., Rorden, C., & Fridriksson, J. (2016). Success of anomia treatment in aphasia is associated with preserved architecture of global and left temporal lobe structural networks. *Neurorehabilitation and Neural Repair, 30*(3), 266—279.

Breier, J. I., Maher, L. M., Novak, B., & Papanicolaou, A. C. (2006). Functional imaging before and after constraint-induced language therapy for aphasia using magnetoencephalography. *Neurocase, 12*(6), 322—331.

Breitenstein, C., Grewe, T., Flöel, A., Ziegler, W., Springer, L., Martus, P., … Bamborschke, S. (2017). Intensive speech and language therapy in patients with chronic aphasia after stroke: A randomised, open-label, blinded-endpoint, controlled trial in a health-care setting. *Lancet, 389*(10078), 1528—1538.

Brownsett, S. L., Warren, J. E., Geranmayeh, F., Woodhead, Z., Leech, R., & Wise, R. J. (2014). Cognitive control and its impact on recovery from aphasic stroke. *Brain: a Journal of Neurology, 137(1)*, 242—254.

Cherney, L. R., & Small, S. L. (2006). Task-dependent changes in brain activation following therapy for nonfluent aphasia: Discussion of two individual cases. *Journal of the International Neuropsychological Society: JINS, 12*(6), 828.

Conroy, P., Sage, K., & Lambon Ralph, M. A. (2009). Errorless and errorful therapy for verb and noun naming in aphasia. *Aphasiology, 23*(11), 1311—1337.

Corbetta, M., & Shulman, G. L. (2002). Control of goal-directed and stimulus-driven attention in the brain. *Nature Reviews Neuroscience, 3*(3), 201—215.

Crinion, J., & Price, C. J. (2005). Right anterior superior temporal activation predicts auditory sentence comprehension following aphasic stroke. *Brain: a Journal of Neurology, 128*(12), 2858—2871.

Crosson, B., Moore, A. B., Gopinath, K., White, K. D., Wierenga, C. E., Gaiefsky, M. E., … Rothi, L. J. G. (2005). Role of the right and left hemispheres in recovery of function during treatment of intention in aphasia. *Journal of Cognitive Neuroscience, 17*(3), 392—406.

Crosson, B., Rodriguez, A. D., Copland, D., Fridriksson, J., Krishnamurthy, L. C., Meinzer, M., … Leff, A. P. (2019). Neuroplasticity and aphasia treatments: New approaches for an old problem. *Journal of Neurology, Neurosurgery, and Psychiatry, 90*(10), 1147—1155.

De Marco, A. T., Wilson, S. M., Rising, K., Rapscak, S. Z., & Beeson, P. M. (2018). The neural substrates of improved phonological processing following successful treatment in a case of phonological alexia and agraphia. *Neurocase, 24*(1), 31—40.

Desikan, R. S., Ségonne, F., Fischl, B., Quinn, B. T., Dickerson, B. C., Blacker, D., … Albert, M. S. (2006). An automated labeling system for subdividing the human cerebral cortex on MRI scans into gyral based regions of interest. *NeuroImage, 31*(3), 968—980.

Eklund, A., Nichols, T. E., & Knutsson, H. (2016). Cluster failure: Why fMRI inferences for spatial extent have inflated false-positive rates. *Proceedings of the National Academy of Sciences, 113*(28), 7900—7905.

Fernandez, B., Cardebat, D., Demonet, J. F., Joseph, P. A., Mazaux, J. M., Barat, M., et al. (2004). Functional MRI follow-up study of language processes in healthy subjects and during recovery in a case of aphasia. *Stroke; a Journal of Cerebral Circulation, 35*(9), 2171—2176.

Fridriksson, J., & Hillis, A. E. (2021). Current approaches to the treatment of post-stroke aphasia. *Journal of Stroke, 23*(2), 183—201.

Fridriksson, J., Morrow-Odom, L., Moser, D., Fridriksson, A., & Baylis, G. (2006). Neural recruitment associated with anomia treatment in aphasia. *NeuroImage, 32*(3), 1403—1412.

Fridriksson, J., Richardson, J. D., Fillmore, P., & Cai, B. (2012). Left hemisphere plasticity and aphasia recovery. *NeuroImage, 60*(2), 854—863.

Friederici, A. D., Meyer, M., & Von Cramon, D. Y. (2000). Auditory language comprehension: An event-related fMRI study on the processing of syntactic and lexical information. *Brain and Language, 74*(2), 289—300.

Fromm, D., MacWhinney, B., & Thompson, C. K. (2020). Automation of the Northwestern Narrative Language Analysis system. *Journal of Speech, Language, and Hearing Research: JSLHR, 63*(6), 1835—1844.

Gainotti, G. (2013). The contribution of language to the right-hemisphere conceptual representations: A selective survey. *Journal of Clinical and Experimental Neuropsychology, 35*(6), 563—572.

Gainotti, G. (2015). Contrasting opinions on the role of the right hemisphere in the recovery of language. A critical survey. *Aphasiology, 29*(9), 1020–1037.

Gajardo-Vidal, A., Lorca-Puls, D. L., Hope, T. M., Parker Jones, O., Seghier, M. L., Prejawa, S., … Price, C. J. (2018). How right hemisphere damage after stroke can impair speech comprehension. *Brain: a Journal of Neurology, 141*(12), 3389–3404.

Geranmayeh, F., Brownsett, S. L. E., & Wise, R. J. S. (2014). Task-induced brain activity in aphasic stroke patients: What is driving recovery? *Brain: a Journal of Neurology, 137(10)*, 2632–2648.

Geranmayeh, F., Chau, T. W., Wise, R. J., Leech, R., & Hampshire, A. (2017). Domain-general subregions of the medial prefrontal cortex contribute to recovery of language after stroke. *Brain: a Journal of Neurology, 140*(7), 1947–1958.

Gilmore, N., Meier, E. L., Johnson, J. P., & Kiran, S. (2019). Nonlinguistic cognitive factors predict treatment-induced recovery in chronic poststroke aphasia. *Archives of Physical Medicine and Rehabilitation, 100*(7), 1251–1258.

Goodman, R. A., & Caramazza, A. (1985). The Johns Hopkins university dysgraphia battery. *Baltimore, MD: Johns Hopkins University*.

Harnish, S. M., & Lundine, J. P. (2015). Nonverbal working memory as a predictor of anomia treatment success. *American Journal of Speech-Language Pathology, 24*(4), S880–S894.

Hartwigsen, G., & Saur, D. (2019). Neuroimaging of stroke recovery from aphasia–Insights into plasticity of the human language network. *NeuroImage, 190*, 14–31.

van Hees, S., McMahon, K., Angwin, A., de Zubicaray, G., & Copland, D. A. (2014). Neural activity associated with semantic versus phonological anomia treatments in aphasia. *Brain and Language, 129*, 47–57.

Heiss, W. D., & Thiel, A. (2006). A proposed regional hierarchy in recovery of post-stroke aphasia. *Brain and Language, 98*(1), 118–123.

Heiss, W. D., Thiel, A., Kessler, J., & Herholz, K. (2003). Disturbance and recovery of language function: Correlates in PET activation studies. *NeuroImage, 20(S1)*, S42–S49.

Higgins, J., Barbieri, E., Wang, X., Mack, J., Caplan, D., Kiran, S., … Parrish, T. (2020). Reliability of BOLD signals in chronic stroke-induced aphasia. *European Journal of Neuroscience, 52*(8), 3963–3978.

Hillis, A. E. (2007). Aphasia: Progress in the last quarter of a century. *Neurology, 69*(2), 200–213.

Hope, T. M., Seghier, M. L., Leff, A. P., & Price, C. J. (2013). Predicting outcome and recovery after stroke with lesions extracted from MRI images. *NeuroImage: Clinical, 2*, 424–433.

Hothorn, T., Bretz, F., & Westfall, P. (2008). Simultaneous inference in general parametric models. *Biometrical Journal: Journal of Mathematical Methods in Biosciences, 50*(3), 346–363.

Hsu, C. J., & Thompson, C. K. (2018). Manual versus automated narrative analysis of agrammatic production patterns: The Northwestern Narrative Language Analysis and Computerized Language Analysis. *Journal of Speech, Language, and Hearing Research: JSLHR, 61*(2), 373–385.

Johnson, J. P., Meier, E. L., Pan, Y., & Kiran, S. (2019). Treatment-related changes in neural activation vary according to treatment response and extent of spared tissue in patients with chronic aphasia. *Cortex; a Journal Devoted To the Study of the Nervous System and Behavior, 121*, 147–168.

Jongman, S. R., Roelofs, A., & Meyer, A. S. (2015). Sustained attention in language production: An individual differences investigation. *The Quarterly Journal of Experimental Psychology: QJEP, 68*(4), 710–730.

Kay, J., Lesser, R., & Coltheart, M. (1992). *PALPA: Psycholinguistic Assessments of Language Processing in Aphasia*. Hove: Lawrence Erlbaum Associates.

Kertesz, A. (2007). *Western Aphasia Battery (Revised). New York, NY: Grune & Stratton*.

Kielar, A., Deschamps, T., Jokel, R., & Meltzer, J. A. (2016). Functional reorganization of language networks for semantics and syntax in chronic stroke: Evidence from MEG. *Human Brain Mapping, 37*(8), 2869–2893.

King, J. W., & Kutas, M. (1995). Who did what and when? Using word-and clause-level ERPs to monitor working memory usage in reading. *Journal of Cognitive Neuroscience, 7(3)*, 376–395.

Kiran, S., Meier, E. L., Kapse, K. J., & Glynn, P. A. (2015). Changes in task-based effective connectivity in language networks following rehabilitation in post-stroke patients with aphasia. *Frontiers in Human Neuroscience, 9*, 316.

Kiran, S., & Thompson, C. K. (2003). The role of semantic complexity in treatment of naming deficits. *Journal of Speech, Language, and Hearing Research: JSLHR, 46*, 773–787.

Kiran, S., & Thompson, C. K. (2019). Neuroplasticity of language networks in aphasia: Advances, updates and future challenges. *Frontiers in Neurology, 10*, 295.

Kleim, J. A., & Jones, T. A. (2008). Principles of experience-dependent neural plasticity: Implications for rehabilitation after brain damage. *Journal of Speech, Language, and Hearing Research: JSLHR, 51*, S225–S239.

Kurland, J., Pulvermüller, F., Silva, N., Burke, K., & Andrianopoulos, M. (2012). Constrained versus unconstrained intensive language therapy in two individuals with chronic, moderate-to-severe aphasia and apraxia of speech: Behavioral and fMRI outcomes. *American Journal of Speech-Language Pathology, 21*, S65–S87.

Kuznetsova, A., Brockhoff, P. B., & Christensen, R. H. B. (2017). lmerTest package: tests in linear mixed effects models. *Journal of Statistical Software, 82*(13), 1–26.

Lambon Ralph, M. A., Snell, C., Fillingham, J. K., Conroy, P., & Sage, K. (2010). Predicting the outcome of anomia therapy for people with aphasia post CVA: both language and cognitive status are key predictors. *Neuropsychological Rehabilitation, 20*(2), 289–305.

Lazar, R. M., Minzer, B., Antoniello, D., Festa, J. R., Krakauer, J. W., & Marshall, R. S. (2010). Improvement in aphasia scores after stroke is well predicted by initial severity. *Stroke; a Journal of Cerebral Circulation, 41*(7), 1485–1488.

Maas, M. B., Lev, M. H., Ay, H., Singhal, A. B., Greer, D. M., Smith, W. S., … Furie, K. L. (2012). The prognosis for aphasia in stroke. *Journal of Stroke and Cerebrovascular Diseases: the Official Journal of National Stroke Association, 21*(5), 350–357.

Marcotte, K., Adrover-Roig, D., Damien, B., de Preaumont, M., Genereux, S., Hubert, M., et al. (2012). Therapy-induced neuroplasticity in chronic aphasia. *Neuropsychologia, 50*(8), 1776–1786.

Meinzer, M., Flaisch, T., Breitenstein, C., Wienbruch, C., Elbert, T., & Rockstroh, B. (2008). Functional re-recruitment of dysfunctional brain areas predicts language recovery in chronic aphasia. *NeuroImage, 39*(4), 2038–2046.

Menke, R., Meinzer, M., Kugel, H., Deppe, M., Baumgärtner, A., Schiffbauer, H., … Breitenstein, C. (2009). Imaging short-and long-term training success in chronic aphasia. *BMC Neuroscience, 10*(1), 1–13.

Moro, A., Tettamanti, M., Perani, D., Donati, C., Cappa, S. F., & Fazio, F. (2001). Syntax and the brain: Disentangling grammar by selective anomalies. *NeuroImage, 13*(1), 110–118.

Musso, M., Weiller, C., Kiebel, S., Müller, S. P., Bülau, P., & Rijntjes, M. (1999). Training-induced brain plasticity in aphasia. *Brain: a Journal of Neurology, 122*(9), 1781–1790.

Nachev, P., Coulthard, E., Jäger, H. R., Kennard, C., & Husain, M. (2008). Enantiomorphic normalization of focally lesioned brains. *NeuroImage, 39*(3), 1215–1226.

Ness, T., & Meltzer-Asscher, A. (2017). Working memory in the processing of long-distance dependencies: Interference and filler maintenance. *Journal of Psycholinguistic Research, 46*, 1353–1365.
Nozari, N., Trueswell, J. C., & Thompson-Schill, S. L. (2016). The interplay of local attraction, context and domain-general cognitive control in activation and suppression of semantic distractors during sentence comprehension. *Psychonomic Bulletin & Review, 23*(6), 1942–1953.
Paulesu, E., McCrory, E., Fazio, F., Menoncello, L., Brunswick, N., Cappa, S. F., … Pesenti, S. (2000). A cultural effect on brain function. *Nature neuroscience, 3*(1), 91–96.
Pedersen, P. M., Vinter, K., & Olsen, T. S. (2004). Aphasia after stroke: Type, severity and prognosis. *Cerebrovascular Diseases, 17*(1), 35–43.
Phillips, C., Kazanina, N., & Abada, S. H. (2005). ERP effects of the processing of syntactic long-distance dependencies. *Cognitive Brain Research, 22(3)*, 407–428.
Plowman, E., Hentz, B., & Ellis, C. (2012). Post-stroke aphasia prognosis: A review of patient-related and stroke-related factors. *Journal of Evaluation in Clinical Practice, 18*(3), 689–694.
Postman-Caucheteux, W. A., Birn, R. M., Pursley, R. H., Butman, J. A., Solomon, J. M., Picchioni, D., … Braun, A. R. (2010). Single-trial fMRI shows contralesional activity linked to overt naming errors in chronic aphasic patients. *Journal of Cognitive Neuroscience, 22*(6), 1299–1318.
Price, C. J. (2012). A review and synthesis of the first 20 years of PET and fMRI studies of heard speech, spoken language and reading. *NeuroImage, 62*(2), 816–847.
Purcell, J., Sebastian, R., Leigh, R., Jarso, S., Davis, C., Posner, J., … Hillis, A. E. (2017). Recovery of orthographic processing after stroke: A longitudinal fMRI study. *Cortex; a Journal Devoted To the Study of the Nervous System and Behavior, 92*, 103–118.
Purcell, J., Turkeltaub, P. E., Eden, G. F., & Rapp, B. (2011). Examining the central and peripheral processes of written word production through meta-analysis. *Frontiers in Psychology, 2*, 239.
Purcell, J. J., Wiley, R. W., & Rapp, B. (2019). Re-learning to be different: Increased neural differentiation supports post-stroke language recovery. *NeuroImage, 202*, Article 116145.
R Core Team. (2022). *R: a language and environment for statistical computing*. Vienna, Austria: R Foundation for Statistical Computing. URL https://www.R-project.org/.
Raboyeau, G., De Boissezon, X., Marie, N., Balduyck, S., Puel, M., Bezy, C., … Cardebat, D. (2008). Right hemisphere activation in recovery from aphasia: Lesion effect or function recruitment? *Neurology, 70*(4), 290–298.
Rochon, E., Leonard, C., Burianova, H., Laird, L., Soros, P., Graham, S., et al. (2010). Neural changes after phonological treatment for anomia: An fMRI study. *Brain and Language, 114*(3), 164–179.
Schwartz, M. F., Saffran, E. M., Fink, R. B., Myers, J. L., & Martin, N. (1994). Mapping therapy: A treatment programme for agrammatism. *Aphasiology, 8*(1), 19–54.
Sebastian, R., & Kiran, S. (2011). Task-modulated neural activation patterns in chronic stroke patients with aphasia. *Aphasiology, 25*(8), 927–951.
Selnes, O. A., Knopman, D. S., Niccum, N., Rubens, A. B., & Larson, D. (1983). Computed tomographic scan correlates of auditory comprehension deficits in aphasia: A prospective recovery study. *Annals of Neurology: Official Journal of the American Neurological Association and the Child Neurology Society, 13*(5), 558–566.
Sestieri, C., Corbetta, M., Spadone, S., Romani, G. L., & Shulman, G. L. (2014). Domain-general signals in the cingulo-opercular network for visuospatial attention and episodic memory. *Journal of Cognitive Neuroscience, 26*(3), 551–568.
Sims, J. A., Kapse, K., Glynn, P., Sandberg, C., Tripodis, Y., & Kiran, S. (2016). The relationships between the amount of spared tissue, percent signal change, and accuracy in semantic processing in aphasia. *Neuropsychologia, 84*, 113–126.
Skipper-Kallal, L. M., Lacey, E. H., Xing, S., & Turkeltaub, P. E. (2017). Right hemisphere remapping of naming functions depends on lesion size and location in poststroke aphasia. *Neural Plasticity, 2017*. https://doi.org/10.1155/2017/8740353
Skipper, J. I., Nusbaum, H. C., & Small, S. L. (2005). Listening to talking faces: Motor cortical activation during speech perception. *NeuroImage, 25*(1), 76–89.
Sliwinska, M. W., Violante, I. R., Wise, R. J., Leech, R., Devlin, J. T., Geranmayeh, F., et al. (2017). Stimulating multiple-demand cortex enhances vocabulary learning. *Journal of Neuroscience, 37*(32), 7606–7618.
Song, X., Wang, X., Alpert, K. A., Wang, L., & Parrish, T. (2015). Robust web-based parallel-optimized minimal preprocessing and analysis pipeline for fMRI big data. *Human Brain Mapping. Honolulu, HI*.
Spell, L. A., Richardson, J. D., Basilakos, A., Stark, B. C., Teklehaimanot, A., Hillis, A. E., et al. (2020). Developing, implementing, and improving assessment and treatment fidelity in clinical aphasia research. *American Journal of Speech-Language Pathology, 29*(1), 286–298.
Stockert, A., Wawrzyniak, M., Klingbeil, J., Wrede, K., Kümmerer, D., Hartwigsen, G., … Saur, D. (2020). Dynamics of language reorganization after left temporo-parietal and frontal stroke. *Brain: a Journal of Neurology, 143*(3), 844–861.
Tao, Y., & Rapp, B. (2020). How functional network connectivity changes as a result of lesion and recovery: An investigation of the network phenotype of stroke. *Cortex; a Journal Devoted To the Study of the Nervous System and Behavior, 131*, 17–41.
Thompson, C. K. (2011). *Northwestern Assessment of Verbs and Sentences (NAVS)*. Evanston, IL: Northwestern University.
Thompson, C. K. (2019). Neurocognitive recovery of sentence processing in aphasia. *Journal of Speech, Language, and Hearing Research: JSLHR, 62*, 3947–3972.
Thompson, C. K., den Ouden, D. B., Bonakdarpour, B., Garibaldi, K., & Parrish, T. B. (2010). Neural plasticity and treatment-induced recovery of sentence processing in agrammatism. *Neuropsychologia, 48*(11), 3211–3227.
Thompson, C. K., & Meltzer-Asscher, A. (2014). Neurocognitive mechanisms of verb argument structure processing. In A. Bachrach, I. Roy, & L. Stockall (Eds.), *Structuring the argument: Multidisciplinary research on verb argument structure* (pp. 141–168). Amsterdam/Philadelphia: John Benjamins Publishing Company.
Thompson, C. K., & Shapiro, L. (2005). Treating agrammatic aphasia within a linguistic framework: Treatment of Underlying Forms. *Aphasiology, 19*(10–11), 1021–1036.
Thompson, C. K., & Weintraub, S. (2014). *Northwestern Naming Battery*. Evanston, IL: Northwestern University.
Turkeltaub, P. E., Messing, S., Norise, C., & Hamilton, R. H. (2011). Are networks for residual language function and recovery consistent across aphasic patients? *Neurology, 76*(20), 1726–1734.
Vigneau, M., Beaucousin, V., Hervé, P. Y., Jobard, G., Petit, L., Crivello, F., … Tzourio-Mazoyer, N. (2011). What is right-hemisphere contribution to phonological, lexico-semantic, and sentence processing?: Insights from a meta-analysis. *NeuroImage, 54*(1), 577–593.
Villard, S., & Kiran, S. (2017). To what extent does attention underlie language in aphasia? *Aphasiology, 31*(10), 1226–1245.
Vincent, J. L., Kahn, I., Snyder, A. Z., Raichle, M. E., & Buckner, R. L. (2008). Evidence for a frontoparietal control system revealed by intrinsic functional connectivity. *Journal of Neurophysiology, 100*(6), 3328–3342.
Walenski, M., Europa, E., Caplan, D., & Thompson, C. K. (2019). Neural networks for sentence comprehension and production:

An ALE-based meta-analysis of neuroimaging studies. *Human Brain Mapping, 40*(8), 2275–2304.

Wanner, E., & Maratsos, M. (1978). An ATN approach to comprehension. In M. Halle, J. Bresnan, & G. A. Miller (Eds.), *Linguistic theory and psychological reality* (pp. 119–161). Cambridge, MA: MIT Press.

Wan, C. Y., Zheng, X., Marchina, S., Norton, A., & Schlaug, G. (2014). Intensive therapy induces contralateral white matter changes in chronic stroke patients with Broca's aphasia. *Brain and Language, 136*, 1–7.

Watila, M. M., & Balarabe, S. A. (2015). Factors predicting post-stroke aphasia recovery. *Journal of the Neurological Sciences, 352*(1–2), 12–18.

Whitney, C., Kirk, M., O'Sullivan, J., Lambon Ralph, M. A., & Jefferies, E. (2012). Executive semantic processing is underpinned by a large-scale neural network: Revealing the contribution of left prefrontal, posterior temporal, and parietal cortex to controlled retrieval and selection using TMS. *Journal of Cognitive Neuroscience, 24*(1), 133–147.

Wierenga, C. E., Maher, L. M., Moore, A. B., White, K. D., Mcgregor, K., Soltysik, D. A., … Briggs, R. W. (2006). Neural substrates of syntactic mapping treatment: An fMRI study of two cases. *Journal of the International Neuropsychological Society: JINS, 12*(1), 132–146.

Wiley, R. W., & Rapp, B. (2019). Statistical analysis in small-N designs: Using linear mixed-effects modeling for evaluating intervention effectiveness. *Aphasiology, 33*(1), 1–30.

Wilson, S. M., Bautista, A., Yen, M., Lauderdale, S., & Eriksson, D. K. (2017). Validity and reliability of four language mapping paradigms. *NeuroImage: Clinical, 16*, 399–408.

Wilson, S. M., Molnar-Szakacs, I., & Iacoboni, M. (2008). Beyond superior temporal cortex: Intersubject correlations in narrative speech comprehension. *Cerebral Cortex, 18*(1), 230–242.

Wright, P., Stamatakis, E. A., & Tyler, L. K. (2012). Differentiating hemispheric contributions to syntax and semantics in patients with left-hemisphere lesions. *Journal of Neuroscience, 32*(24), 8149–8157.

Yarkoni, T., Speer, N. K., & Zacks, J. M. (2008). Neural substrates of narrative comprehension and memory. *NeuroImage, 41*(4), 1408–1425.

Zaidel, E. (1976). Auditory vocabulary of the right hemisphere following brain bisection or hemidecortication. *Cortex; a Journal Devoted To the Study of the Nervous System and Behavior, 12*(3), 191–211.

Zatorre, R. J., Meyer, E., Gjedde, A., & Evans, A. C. (1996). PET studies of phonetic processing of speech: Review, replication, and reanalysis. *Cerebral Cortex, 6*(1), 21–30.

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